

Greener Pastures? A Labor Market Theory of Climate Governance

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Firm interest in climate change has rapidly mounted. We argue that a consequent public-private competition for climate expertise undermines public climate governance. Governments seek climate experts to develop and implement climate policies. Firms seek similar experts to affect policy choice, skirt enforcement, and measure climate risks to commercial interests. As firm interest in climate grows, the government's ability to attract experts declines, eroding its capacity for climate governance. To support this argument, we measure private sector demand for experts with unique data on U.S. job postings, paired with administrative data on government bureaucrats and metrics of the quality of climate governance. Our findings offer a novel labor-based account of global climate politics, contributing to the study of labor markets in political science, the sources of bureaucratic expertise, and the net effects of corporate social responsibility.

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Private sector attention to climate change has deepened and broadened in recent years. The “existential” threat of decarbonization has mobilized a range of polluting firms (Colgan, Green, and Hale 2021; Cory, Lerner, and Osgood 2021); fossil fuel industry representatives outnumbered nearly all country delegations at the most recent UN climate conference.¹ Risks of unmitigated climate change have driven other firms to invest in bolstering their climate resilience (Condon 2023), while new regulations — such as emissions reporting requirements in California and carbon border adjustment mechanisms in Europe — increasingly incentivize firms to measure and improve their climate performance.² Wrote the chief executive of BlackRock in 2020, “There is no denying the direction we are heading. Every government, company, and shareholder must confront climate change.”³

One effect of ascendant firm interest in climate is intensified demand for climate *expertise*. While the causes of climate change are well known, designing, lobbying over, and managing policies meant to mitigate warming are technically intensive tasks (Stokes 2020; Hai 2024). Models of climate impacts, which guide private and public investments in adaptation and resilience, are difficult to develop and require “tremendous amounts of information” that are often opaque to lay audiences (Gazmararian and Tingley 2023, 65). Climate action across both the private and public sectors accordingly necessitates a high level of expert knowledge. Yet both firms and governments often lack the personnel able to effectively provide this (Condon 2023). While the supply of climate expertise is increasing, that growth has to date been outstripped by growth in demand, complicating the recruitment of climate specialists by both firms and governments (Committeri et al. 2022, 10).⁴

We argue that mobilization on climate by firms curtails states’ capacity for climate governance and in turn reduces the quality of that governance. The mechanism underlying this is a transna-

¹*The Guardian* 2025 [perma.cc/N9XR-N39N].

²See, e.g., PwC 2026 [bit.ly/3NVIgrI].

³*BlackRock* 2020 [perma.cc/6X63-WSG5].

⁴*Business Insider* 2022 [perma.cc/56PE-Y4Z4], *CLEW* 2023 [perma.cc/VV9C-TX54].

tional labor market competition between governments and firms for high-quality climate experts. Policymakers and regulators seek experts to design, interpret, and enforce climate policies. Firms demand those same experts to strengthen their lobbying over those policies and to meet legal and fiduciary obligations, such as to measure carbon emissions or physical climate risks to commercial interests. We characterize this competition in two steps. First, we argue that as firm demand for climate experts increases, the relative quality of expertise in government declines. Bureaucracies struggle to counter the remunerative advantages of for-profit employers, impeding their ability to recruit and retain in-demand talent. Second, as a consequence of this, we propose that firm demand reduces the quality of public climate governance; specifically, we suggest that it undermines government production of climate information and its enforcement of climate regulations.

To test this argument, we use new measures of public-private competition for expertise in the case of the United States, as well as granular data on characteristics of U.S. federal bureaucrats and indicators of the quality of public climate governance. For the former, we leverage unique microdata on the near universe of jobs advertised by firms and the federal government over the last decade. These data reveal a rush of demand for climate experts across both domestic and foreign firms: the rate of job postings in the 2020s is triple that of the 2010s, and the private sector wage premium for experts has likewise surged, indicative of mounting competition for their services. Noting the likely variation in the motivations of hiring firms, we use firm-level climate performance metrics from a prominent advocacy group to distinguish between firms with leading and lagging records on climate and decompose the growth in demand accordingly. We find that top-ranking firms, which we expect to be more committed to pro-climate objectives, account for less than a quarter of demand over the 2015–24 period.

Acknowledging the possibility that public and private employers seek different forms of climate expertise, we use granular data on demand for specific occupations and skills to demonstrate that the types of experts sought by firms overlap substantially with those sought by government. Between 2011 and 2024, roughly half of all private sector climate job postings targeted the same types of climate workers recently pursued by federal agencies. Likewise, we estimate that the

average level of private sector competition for climate-related hiring by federal employers — the average number of firm postings overlapping with a single simultaneous government posting — more than doubled between these years.

Pairing these hiring metrics with individual-level data on federal civil servants, we identify that growth in private sector demand meaningfully affects bureaucratic composition. It reduces average rates of educational attainment among climate-related bureaucrats, particularly those at the highest ranks of the civil service. Evidence suggests that a key mechanism is poaching of climate bureaucrats by firms: firm demand significantly increases the number of such staff who choose to leave the federal government. While federal agencies appear able to compensate for these quits with new hires at lower ranks of the bureaucracy, we document a net outflow of talent at more advanced levels, consistent with the greater private sector wage premiums that exist for higher skilled and more senior workers. We find that these outflows are driven by demand from both climate-leading and climate-lagging firms, suggesting that government expertise is partly being reallocated towards firms with weaker records on climate.

To identify the implications for downstream climate governance, we measure changes in the quality of government-produced climate research and in the efficiency of climate-related regulatory enforcement. We first find that firm demand for climate experts reduces citations to subsequently published government climate research — compared to climate research by non-government authors, as well as government research on non-climate topics — which we interpret as evidence of decline in the quality of that work. We then find that firm demand slows regulatory enforcement, measured with data on the number of industrial facilities with “unresolved” violations of federal climate-relevant regulations.

These findings build our understanding of climate, firm, and bureaucratic politics. Our labor market-based conception of climate governance diverges from the growing literature that emphasizes electoral and lobbying (dis)incentives for climate action (e.g., Kennard 2020; Gaikwad, Genovesi, and Tingley 2022). While we agree on the importance of these factors, the labor market phenomena we identify may bind governments even in otherwise favorable political settings. Our

findings connect well with work instead approaching climate as a complex issue marked by public-private expertise gaps (Green 2013). Building on literature that analyzes bureaucratic behavior and regulatory outcomes under these conditions (McCarty 2017; Perlman 2023), we offer a dynamic understanding of how the allocation of expertise across sectors evolves. In doing so, we illustrate how states’ ability to draw on epistemic communities for policymaking support — a relationship long recognized as critical to global climate governance (Haas 1992) — varies over time.

The paper moreover clarifies the consequences of spread in corporate social responsibility (CSR) and environmental, social, and governance (ESG) initiatives. Scholars have been wary of CSR’s ability to mitigate climate change due to the concern that firms employ these initiatives in bad faith: rather than truly attempting to fight climate change, private governance can be used to preempt public regulation (Malhotra, Monin, and Tomz 2019) or mask the lack of substantive mitigation efforts (Berliner and Prakash 2015). This paper suggests that such initiatives, even when adopted in good faith, have the potential to compromise public regulatory capacity. All such contributions advance the burgeoning literature on how firms shape climate policy trajectories (e.g., Green et al. 2022; Bayer 2023; Gazmararian and Milner 2026; Finnegan and Meckling 2026).

Finally, our paper contributes a new account of how bureaucracies develop, retain, and lose expertise in novel policy domains. Scholars emphasize determinants of expertise acquisition and recruitment that are internal to government, such as delegation of discretion by political principals or changes in messaging strategies (Gailmard and Patty 2007; Ashraf et al. 2020; Clark and Zucker 2024). We instead contend that consumption of expertise by firms impedes its acquisition in the short term by government. By arguing that public and private institutions draw from the same pool of high-skilled workers—conceiving of bureaucracies in terms of their place in broader labor markets—we reframe expertise as a rivalrous good. While we believe this argument is particularly relevant to climate, it may extend to other emergent and complex issue areas like artificial intelligence, where there is likewise surging global demand for and limited supply of expertise.

ALLOCATION OF EXPERTISE

Our theory understands climate as a novel political issue, one in which most public and private institutions initially lack expertise. Political institutions dedicated to climate governance began to emerge in the 1980s and 1990s, often closely tied to communities of scientific experts (Haas 1992). Attention to climate at generalist institutions, however — those without mandates specific to climate — took longer to emerge. Interest in climate at international financial institutions, for instance, did not seriously mount until the mid-2010s (Clark and Zucker 2024). Central banks and financial regulators were further delayed in embracing the issue (Quorning 2023). Industry interest in climate is likewise recently ascendant (Lerner and Osgood 2023).

Climate is moreover distinct in its technical complexity. Take one task of rapidly growing interest across the public and private sectors: modeling of future climate-induced physical risks to economic assets. Firms, government agencies, and international organizations use increasingly intricate catastrophe models to measure such risks to commercial interests and government budgets, as well as to plan climate adaptation programs. Development of these models can require “a team of 10–20 leading experts from a variety of disciplines (specialists with PhD degrees) . . . work[ing] together for over two to three years.”⁵ As another example, states increasingly seek to evaluate financial institutions’ resilience to climate stressors. Efforts to conduct such tests have been plagued by a broad “lack of technical expertise on climate science and environmental economics” at both firms and regulators.⁶ Measurement of emissions, development of net-zero pathways, and identification of low-carbon investment opportunities likewise require expert involvement (Carnegie, Clark, and Zucker 2024).⁷ Government officials cite technical issues relating to data and measurement as barriers to climate policymaking (Gazmararian and Tingley 2023, 65).

⁵*Geneva Association* 2018 [perma.cc/L22V-DPWD], p. 25.

⁶*Network for Greening the Financial System* 2020 [perma.cc/JN74-ZHDM], p. 7.

⁷Also see *FT* 2023 [perma.cc/V3FN-42S6].

Demand for Climate Expertise

The novelty and complexity of climate change create incentives to acquire climate expertise. A large literature recognizes the importance of bureaucratic expertise to effective domestic and international governance (e.g., Abbott and Snidal 1998; Lewis 2007). In environmental and scientific contexts, academic expertise has long factored prominently into the development of international agreements, domestic policy, and regulatory action (Haas 1992; Perlman 2023; Hai 2024). We accordingly assume that governments seek expertise to make and enforce climate-related policies and regulations.

Public institutions' push for expertise is likely most intense when political leaders want to design and enact ambitious climate policies. But it should endure after those leaders are replaced by figures with weaker climate preferences or explicitly anti-climate platforms. To the extent that civil servants remain motivated by policy outcomes despite changes in political leadership, they may still seek expertise to implement and enforce existing statutes when they retain that delegated authority across leadership transitions. The ascension of anti-climate leaders, including populists who reject technocratic expertise, may dampen demand in bureaucracies under their control (Carnegie, Clark, and Zucker 2024),⁸ but not necessarily at the subnational level or in offices associated with opposition politicians. It is also important to distinguish between expertise and political preferences; individuals with deep technical knowledge may be ideologically flexible and willing to carry out anti-climate agendas (see Gailmard and Patty 2007).

Profit-motivated firms in the private sector likewise benefit from acquiring expertise. Firms seek expertise in large part to identify and manage risks to commercial interests posed by decarbonization and physical climate impacts (Gazmararian and Milner 2026). Climate expertise further allows firms to meet consumer demand for new products; banks, for example, are issuing "climate

⁸In some cases, such as the second Trump administration, right-wing populist leaders actively seek to remove climate experts (*NPR* 2025, perma.cc/JU5C-XS8F). The assumption that governments demand expertise may not hold in such instances.

communiques” with scientific data to satisfy “clients’ demand for [climate] information.”⁹ Financial institutions invest in modeling how future climate disasters will stress their portfolios (Condon 2023). Agricultural firms have been pressed to assess how climate change will affect production. Likewise, attention to net-zero transition plans, carbon emissions accounting, and ESG has increased demand for expertise in corporate sustainability (Lerner and Osgood 2023).¹⁰ Firms may also expand their ranks to satisfy stakeholder demands for climate action (Thrall 2021).

Expertise also serves as a source of policy influence. In complex issue areas like climate, private sector command of issue-specific expertise confers sway over less-informed policymakers and regulators (McCarty 2017; Brutger 2024). Perlman (2023), for example, illustrates how firms strategically withhold private information about their operations to manipulate rulemaking. This is a salient concern; information about the physical vulnerability and carbon intensity of assets and supply chains is often held privately by firms, especially in the absence of binding disclosure requirements. Firms may use such expertise to persuade policymakers or, consistent with Hall and Deardorff (2006), support the work of like-minded but less knowledgeable policymakers.

Expertise further helps firms comply with regulations and guard against unfavorable government actions. To the extent that expertise supports technological innovation and corporate flexibility, carbon-intensive firms with more in-house expertise should be better positioned to adapt to strict climate regulations (Kennard 2020).¹¹ More sophisticated firms more capably avert regulatory scrutiny (Hall and Miler 2008) and quell litigation risks (de Figueiredo and de Figueiredo 2002). As Green (2013) highlights, firm command of expertise may in fact prompt states to delegate rule-making authority to private actors.

As this discussion suggests, we do not claim that private sector demand is unaffected by government policy. Greater climate policy ambition plausibly augments firm interest in experts for

⁹*Bloomberg* 2025 [bit.ly/4gKPYNJ].

¹⁰*Agri-Pulse* 2020 [perma.cc/7MVW-GWPM]; *GreenBiz* 2022 [perma.cc/5SQ3-9E2N].

¹¹Green et al. 2022 find that R&D-intensive firms are typically more supportive of decarbonization, “reflecting the idea that more ‘innovative’ firms are more able to shift strategies” (2057).

regulatory compliance and lobbying purposes. While we do not hypothesize about or test for the sources of firm demand in this paper, we note that policy-related drivers of demand would be analytically problematic if they also affect the composition and behavior of government bureaucracies. We address this possibility in the following theoretical discussion and empirical analysis (see, e.g., “Alternative Explanation” on page 23).

Balance of Climate Expertise

Demand across governments and firms generates a market for *climate experts*: individuals with advanced technical knowledge of climate change and decarbonization. Here we group together individuals who acquire climate-related knowledge via formal schooling and via professional experiences. The former category encompasses researchers with advanced technical training related to climate across engineering, law, and the natural and social sciences. The latter camp includes government bureaucrats who learn about climate while on the job and private sector employees who do the same.

The novelty of climate as a hiring priority explains the scarcity of climate experts available to public and private employers. For employees of public institutions or firms, there is less incentive and opportunity to invest in acquiring expertise when climate is not a focus of political principals, firm executives, or prospective employers. There should also be less production of academic expertise; the logic of Mincer (1974) implies that individuals should have invested less in learning climate skills when labor market demand was lower.

A critical premise of our theory regards the elasticity of labor supply: the rate at which the supply of climate experts grows to meet rising demand. We argue that, while increasing demand for climate expertise is sure to generate more climate experts, it will only do so after a delay. This is consistent with work on human capital; when hiring and wage offers increase for workers with certain skill sets, individuals lag in updating their education decisions (Deming and Noray 2020). Supply may particularly lag demand in the climate field; climate change is new as a major topic of academic study, with limited course options at many schools (Hess and Collins 2018), and politicization of the issue may create uncertainty around future demand for climate skills.

There is empirical support for this intuition. A recent LinkedIn study concluded that while “the supply of green talent has ... grown” across countries in Asia, Latin America, and Europe, it consistently “lags demand.”¹² A survey found that while most energy firms intend to “hire skilled workers for the energy transition,” they have “faced difficulties finding qualified applicants for almost all occupation categories.”¹³ Sustainability executives have cited “difficulty in hiring talent with climate change skills” as a primary impediment to decarbonization.¹⁴ Efforts to hire new climate specialists at international financial institutions have been complicated by a “limited number” of experts available worldwide (Committeri et al. 2022, 10). While supply of climate expertise may eventually equalize with demand — a possibility we discuss in the conclusion — short-term shortages following demand shocks are theoretically plausible and empirically apparent.

Bureaucratic Composition

We first theorize that private sector demand affects bureaucratic composition and quality (Appendix A contains a simple formalization of the theory). We argue that as demand rises across employers, government agencies will see erosion in the quality of their in-house climate expertise. The assumption underlying this claim is that firms can offer experts higher wages than public institutions, as ample empirical work shows. Research points to a consistent private sector wage premium for the highest skilled workers in the U.S. and other high-income democracies, with government employers reacting sluggishly to private sector wage growth (Katz and Krueger 1991; Abdallah, Coady, and Jirasavetakul 2023). Government wage offers are often constrained by salary schedules that determine the maximum compensation for given positions or career grades. We formalize this insight in Appendix A by treating government compensation as inelastic to experts’ skill past a certain wage level (whereas private sector compensation faces no similar constraint).

We assume, further, that experts primarily select career paths according to the wages that they are offered. All else equal, experts will prefer to take higher-paying jobs; while public agencies can

¹²LinkedIn 2024 [perma.cc/B2QT-DM2K].

¹³IEA 2024 [perma.cc/MFF9-AZ4H].

¹⁴EY 2023 [perma.cc/VR5Y-8FUC].

offer superior job security via civil service protections, scholarship indicates that pay imbalances between these institutions and private employers still entice transitions towards the latter, hence our focus on wages (Shepherd and You 2020). We hasten to note that wages are not the sole motivation behind job choice; experts may also possess some pro-social public service motivation, or ideological commitment to working on climate policy. Pro-social and policy motivations of this sort would attenuate, but not eliminate, wage motivations.¹⁵

Growth in private sector demand for climate expertise should produce higher wage offers to scarce experts, as firms seek to out-compete each other and the government for their labor. To the extent that wage offers are scaled to expert *quality*, government agencies should remain able to compete for lower quality experts in periods of mounting private sector demand.¹⁶ But the highest quality experts — those commanding the most lucrative wage offers from firms — should be increasingly difficult for government agencies to attract as firm demand mounts, with rigid budget constraints and pay schemes limiting their ability to flexibly negotiate with high-end experts. Insofar as pro-social motivations are distributed across climate experts independent of their quality (i.e., higher quality experts are not also significantly more pro-social), we should still expect a net loss of high-quality climate experts from government agencies.¹⁷ This implies the following:

Hypothesis 1. *Growth in private sector demand for climate expertise reduces the average quality of climate-relevant bureaucrats in government, by limiting (a) the retention and (b) the recruitment of high-quality bureaucrats.*

Previously we noted that climate policy ambition may, alongside firm demand, affect the makeup of government bureaucracies. The effect of government interest in climate policy on bureaucratic composition would likely be inverse that of firm demand. Firm demand should reduce

¹⁵The formalization in Appendix A incorporates such non-wage motivations.

¹⁶For this reason, we do not predict that firm demand will produce an absolute reduction in the number of climate experts in government. We rather predict a reduction in the *average* quality of such experts in government.

¹⁷Prior work finds that bureaucrat quality and pro-sociality vary independently (Ashraf et al. 2020).

the quality of climate bureaucracies; conversely, pro-climate governments presumably invest in *greater* bureaucratic capacity to better design and implement climate policies, particularly where high-quality bureaucrats and the government have similar climate policy objectives.¹⁸

Quality of Climate Governance

Our theory predicts that as private sector demand for climate expertise increases, the highest quality experts will increasingly opt to work for firms rather than government. To the extent that bureaucratic expertise supports policy choice and implementation (Lewis 2007), the loss of high-quality climate bureaucrats should constrain the ultimate quality of public climate governance. This is likely to be true even if governmental climate expertise declines only *relative* to the climate expertise available in the private sector. Given that much of the work of government climate experts entails regulating firms, and assuming that many firms prefer less regulation than the government finds socially optimal, aggregate shifts in the balance of expertise towards firms are likely to make the task of regulation more difficult, even if there is no absolute decline in the quality of the climate bureaucracy (see, e.g., Perlman 2023). While substantial work on informational and subsidy-based lobbying has demonstrated that firms are often willing to share their expertise with policymakers, this is conditional on alignment of interests (Hall and Deardorff 2006; Kennard 2020; Thrall 2025); firms are unlikely to voluntarily disclose information that would negatively affect them. As noted above, some firms may use climate experts to improve their climate performance or regulatory compliance. Hiring by those firms may have positive effects on climate outcomes. We assume that on average, however, privately employed climate experts will work towards profit-seeking goals, which may or may not align with the public interest.¹⁹

¹⁸Erosion in the quality of climate bureaucracies is most apparent under climate-skeptical populist regimes (Carnegie, Clark, and Zucker 2024).

¹⁹For example, a former EPA administrator who became an energy industry lobbyist subsequently served as an expert witness for a coal company in Clean Air Act-related litigation against the EPA (*E&E News by Politico* 2016, perma.cc/SUR8-C9JM). A former chief scientist at the National Oceanic and Atmospheric Administration now analyzes climate risks to investments for

We focus on two dimensions of climate governance: government production of climate information and its enforcement of climate regulations. Global climate governance is dependent upon climate data produced by governments, such as emissions inventories (Carnegie, Clark, and Zucker 2024), and research by government scientists routinely factors into climate policymaking, such as decisions over adaptation investments.²⁰ Global climate action is likewise dependent upon effective bureaucratic enforcement of climate laws enacted by political principals. Scholars document significant variation in the extent to which climate regulations are effectively enforced (Konisky and Woods 2010); more generally, work holds that the quality of policy implementation is a function of the capacity of participating bureaucrats (Gailmard and Patty 2012). The loss of bureaucrats to firms and hampered recruitment should limit both government production of climate information and its enforcement of climate regulations.

Hypothesis 2. *Growth in private sector demand for climate expertise reduces the quality of public climate governance.*

EXTENT OF PUBLIC-PRIVATE COMPETITION

We focus empirically on the United States, though see the theory as applicable across countries where there are private sector wage premia for high-skilled workers.²¹ The U.S. features a large administrative state and, like other advanced economies, has experienced a high level of private sector climate hiring in recent years.²² We first characterize the nature of public-private competition for climate experts and profile the firms that have sought to hire these experts. We then test our theory, linking intensification of this competition — increased firm demand for workers also

JPMorgan (*Bloomberg* 2026, perma.cc/Q7GY-YSR4).

²⁰*New York Times* 2025 [perma.cc/BH4X-KT9Y].

²¹This is most common in high-income countries (Abdallah, Coady, and Jirasavetakul 2023).

²²On global demand for climate expertise, see *FT* 2025 [perma.cc/43GP-SV92]. Demand has grown at international organizations (Committeri et al. 2022), non-governmental organizations (*Devex* 2026, perma.cc/6E53-AJA8), and in academia (*Axios* 2024, perma.cc/L4PL-4QKE).

sought by government — to the quality of the U.S. federal bureaucracy and in turn the quality of government climate research and regulatory enforcement.

We measure demand for climate experts using data on the near universe of job postings in the U.S. between 2010–2024, covering 434 million jobs across the private and public sectors. We draw these data from Lightcast, a labor market analytics company that collects large amounts of information on individual job openings, including job descriptions, education requirements, requested skills, and salary ranges. Despite its unique breadth and detail, Lightcast data have not been widely used in political science nor in any studies of climate change to our knowledge.²³

We identify full-time job postings targeted at climate experts via a keyword search of job descriptions. We develop a list of thirty terms, based on skills and tasks commonly involved in climate work. Terms include, for example, “climate change,” “climatolog[y],” and “meteorolog[y].” Jobs are classified as climate related if their description includes at least one term in this dictionary. This correlates closely with an alternative measure of climate-related job postings, based on Lightcast codings of the skills requested for individual jobs. Appendix C details these measures.

Our theory concerns competition between firms and governments over a common pool of climate experts. In practice, however, the private and public sectors may seek different types of climate experts. In secondary analyses, we consider private sector demand for types of climate workers that are concurrently pursued by the federal government. To identify such workers, we extract Lightcast data on the occupations targeted in individual job postings. We code a private sector job as *overlapping* with federal demand when a federal agency sought to hire in that same occupational category in the preceding two years.²⁴

Figure 1 summarizes changes in demand for climate experts. Affirming our theoretical intu-

²³Data from Lightcast, formerly known as Burning Glass Technologies, have been used in labor economics (Deming and Noray 2020).

²⁴We use six-digit occupational codes from the Standard Occupational Classification System. We define federal postings as jobs listed on usajobs.gov, the main application portal for federal jobs; we identify some additional federal jobs through a manual review of employer names.

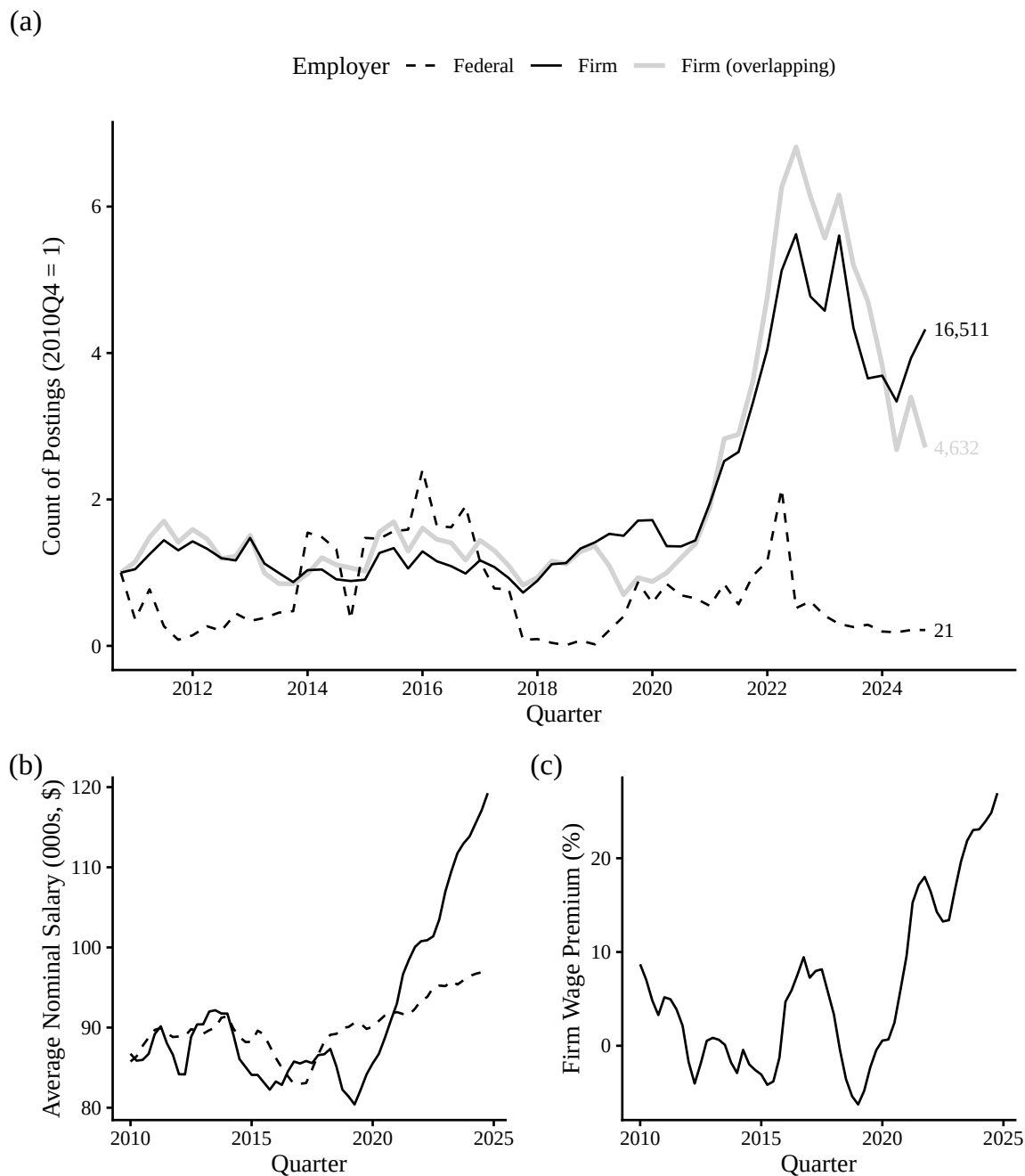


Figure 1: Panel (a) plots counts of climate-related job postings by the federal government and firms, indexed to 2010Q4; the gray line depicts the number of firm postings targeting the same occupations as federal postings by quarter. Annotations to the right of each line indicate posting count in 2024Q4. Panel (b) plots average nominal salaries across firms (solid) and federal agencies (dashed); eight-quarter rolling means. Panel (c) plots private sector wage premiums, after matching jobs by occupational code and minimum educational requirement; eight-quarter rolling means.

itions, two trends are immediately apparent. First, panel (a) illustrates a surge in private sector demand for climate expertise in recent years. Firms advertised, on average, 37,533 climate-related jobs in each quarter between 2021 and 2024, more than three times the rate of hiring in the 2010s.²⁵ Firm job postings increasingly targeted the same types of workers as federal postings as well, pointing to intensified public-private hiring competitions. In the fourth quarter of 2024, each federal climate-related job posting overlapped on average with 129 contemporaneous postings by firms, a 73% increase over 2021 and elevenfold increase over 2016. About 56% of postings by firms overlapped with recent federal postings.

Second, this proliferation of climate job postings mirrors growth in the wages on offer. Panel (b) indicates that while average nominal salaries in the private sector remained roughly steady throughout the 2010s, they have dramatically increased since 2019. While federal pay has also increased, it has lagged well behind, as indicated by the growing private sector wage premium depicted in panel (c). To estimate the private sector wage premium, we matched federal jobs to firm jobs on the basis of minimum educational requirements and six-digit occupational codes from the U.S. government’s Standard Occupational Classification System. Panel (c) documents that federal and firm pay was near parity throughout the 2010s. Between 2019 and 2024, however, the private sector wage premium has increased by about 30 percentage points.

Profile of Hiring Firms

Which types of firms hire climate experts? Professional, scientific, and technical services firms (NAICS 541) are the most frequent advertisers of climate-related jobs in the data. In 2015, for example, the infrastructure consultancy AECOM advertised for an environmental engineer who would “evaluate and determine sources and methods of controlling pollutants in air,” “provide

²⁵This elevated level of hiring persisted after 2024; postings from 2025Q1–2026Q2 stood above the 2024Q4 level. We exclude post-2024 quarters from the analysis due to the unique efforts of the second Trump administration to downsize the federal bureaucracy (*Washington Post* 2025 wapo.st/4fiGla4); our theory focuses on cases where government agencies seek climate expertise.

technical-level support for ... litigation,” and assess regulatory compliance. Carbon-intensive companies are also active in climate-related hiring. Utilities companies (NAICS 221) are among the most frequent advertisers of climate-related jobs in our data. Southern California Edison, for example, sought in late 2023 a senior advisor to help incorporate climate adaptation into project planning, offering pay up to \$218,000 — \$34,500 more than the maximum salary available to most federal workers that year.²⁶ On average, hiring firms are more likely than non-hiring firms in the same industries to lobby and contract for the government (Appendix D).

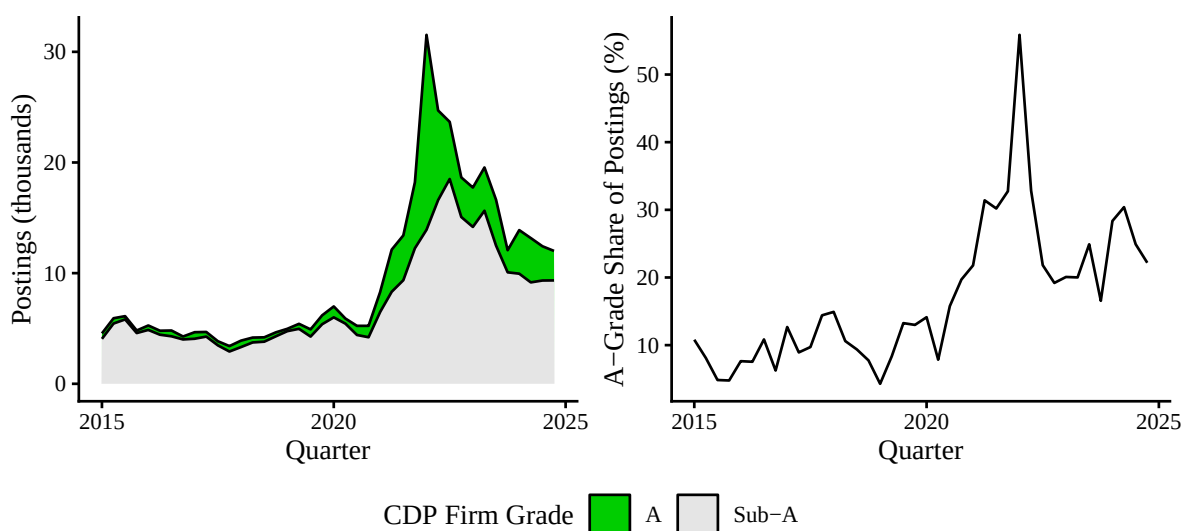


Figure 2: Left panel plots quarterly counts of climate-related job postings by firms that received an A grade or less than an A grade from CDP. Right panel plots the share of jobs posted by A-grade firms by quarter.

Some motivations behind hiring of climate experts are consistent with global climate objectives, such as efforts to reduce firm-level emissions. Others, such as an interest in bolstering lobbying capacity or averting regulatory scrutiny, have ambiguous or negative implications for progress towards those goals. To distinguish between these firms, we draw on grades of corporate climate performance published by CDP, formerly known as the Carbon Disclosure Project (see Appendix I for details). Figure 2 depicts demand for climate experts over time, disaggregating between firms

²⁶In 2023, the highest salary for federal workers based in Washington, D.C., and on the General Schedule pay scale was \$183,500 [federalpay.org/gs/calculator].

that received an A grade from CDP, indicating top climate performance, and less than an A grade.²⁷ As illustrated here, the bulk of demand has come from firms with lagging performance on climate (less than an A grade). Just 23% of climate job postings between 2015–24 came from firms with A-grade climate records. This proportion has grown over time, though as of 2024Q4 postings from sub-A grade firms still outnumbered A-grade postings three-to-one.

BUREAUCRATIC COMPOSITION

How does private sector demand for climate expertise affect the composition of the federal climate bureaucracy? To measure climate expertise in the public sector, we draw on FedScope data from the U.S. Office of Personnel Management (OPM) for the years 2010–2024. FedScope data provide individual-level employment information on 96% of federal employees at the quarterly level. They detail the education, professional grades, and occupations of federal workers. These data, limited to college-educated workers, contain 50.2 million bureaucrat-quarter observations.²⁸

We use occupation codes to determine which bureaucrats hold climate-relevant positions. We identify thirteen climate-related occupations based on a manual review of the available codes and of the occupations sought in federal climate-related job postings (Appendix E). These occupations include individuals who work on scientific aspects of climate change, as well as those focused on regulatory compliance and enforcement, such as environmental protection specialists.²⁹ These codes thus plausibly encompass the climate experts sought by firms. Fewer than 5% of bureaucrat-quarter observations are coded as climate relevant. Appendix F reports summary statistics.

We aggregate these data in two ways, in both cases filtering to the twenty federal agencies with at least 1,000 climate-relevant employee-quarter-year observations (see Appendix F). Our

²⁷We focus on grades in 2014 and outcomes for 2015–2024 given the possibility that climate hiring affects grades. Firms not contacted by CDP are excluded from these analyses.

²⁸The data do not include identifiers that permit matching of bureaucrats across waves.

²⁹On the role of environmental protection specialists, see OPM [perma.cc/NJ92-ZPS7]. Not everyone in these occupations works specifically on climate, but many do (*Union of Concerned Scientists* 2021, perma.cc/46NV-63F2); FedScope data do not permit more precise coding.

first approach aggregates to the agency-quarter-year-job type (climate relevant or not) level. For example, we have separate observations for Department of Agriculture, 2018Q2, climate-relevant, and Department of Agriculture, 2018Q2, other (non-climate relevant). This structure allows us to use each agency's non-climate relevant staff as a comparison group for its climate experts; non-climate employees should be unaffected by private sector climate demand, while accounting for agency-specific hiring practices.

Our second approach accounts for the General Schedule (GS) system that structures the federal civil service. Most federal civilian workers are classified at a pay grade ranging from GS-1 (most junior) to GS-15 (most senior). These grades reflect the “level of difficulty, responsibility, and qualifications required” for a position; grades from GS-1 through GS-7 are considered entry level, GS-8 through GS-12 mid-level, and GS-13 through GS-15 upper level and supervisory.³⁰ Approximately 70% of climate-relevant bureaucrats are classified at or above GS-11. In this approach, we aggregate data to the agency-quarter-year-*GS grade*-job type level, allowing us to compare staff within the same GS grade.

This within-GS grade approach addresses the possibility that climate policymaking confounds the relationship between firm demand and bureaucratic composition. Pro-climate administrations may trigger increased firm demand for expertise and simultaneously expand the lower ranks of the climate bureaucracy with less educated staff, as they seek more staff to do the day-to-day work of climate policy implementation. GS grades, usefully, capture the administration's hiring priorities. As part of the annual appropriations process, federal agencies submit staffing plans and funding requests that specify the number and level of positions they intend to fill; once approved, jobs are advertised at these pre-authorized GS grades. Accordingly, by comparing staff within the same GS grade, we effectively control for shifts in administrative hiring priorities.

³⁰OPM [perma.cc/YG7N-BKY9]; FederalPay.org [federalpay.org/gs/2025/GS-13].

Bureaucratic Quality

We measure bureaucratic quality as the share of bureaucrats who hold a graduate degree.³¹ Our identification strategy approximates a difference-in-differences design with continuous treatment (Callaway, Goodman-Bacon, and Sant’Anna 2026); our desired estimand is thus the average treatment effect on the treated (see Appendix M for further discussion). The treatment, plotted in Figure 1(a), is the logged count of climate-relevant private sector job postings in a given quarter; in secondary tests, we re-specify this as the count of firm job postings overlapping with federal demand. In tests pooling over GS grades, the treated group is the population of climate-relevant employees within each federal agency in our sample; the comparison group is the population of non-climate-relevant employees within those agencies. We estimate the following by ordinary least squares:

$$Y_{atc} = \delta [\text{demand}_{t-1} \times \mathbb{I}(c = \text{climate})] + \zeta \mathbf{Z}_{tc} + \phi_a + \eta_t + \varepsilon_{atc}$$

where a indexes agencies, t quarter-years, and c the climate relevance of the job type. All models include quarter-year fixed effects with standard errors clustered by agency.³² In tests disaggregating bureaucrats by GS grade, we include agency-GS grade fixed effects in lieu of agency fixed effects. Our coefficient of interest is δ , the interaction between private sector climate demand and a binary indicator for climate-relevant employees.

One possibility is that any association between firm demand and bureaucratic quality owes not to changes in demand for *climate* skills, but rather broad shifts in high-skilled labor markets with which demand for climate expertise may be correlated; this would pose a threat to identification

³¹We do not claim that a graduate degree is necessary for climate work. Climate governance involves complex analytical work and specialized skills distinct from those typically held by bureaucrats (Stokes 2020). We thus assume that graduate training, a correlate of human capital, proxies for ability to perform climate-related tasks. The U.S. Department of Labor O*NET listing for climate change policy analysts, for example, states that “most of these occupations require graduate school” [perma.cc/27XS-5ZSX].

³²Results are consistent when alternatively clustering by quarter-year.

by violating the parallel trends assumption. To account for this, we control for the interaction of climate relevance with quarterly firm demand for non-climate-related software workers (logged count of postings), a proxy for general demand for technically sophisticated workers.³³

DV:	% Staff With Graduate Degree			
	<i>GS Pooled</i>	<i>GS: Mid Level</i>	<i>GS: Upper Level</i>	
	(1)	(2)	(3)	(4)
Demand × Climate-relevant	−1.233*** (0.313)		−1.381 (1.574)	−2.747* (0.975)
Demand (overlapping) × Climate-relevant		−0.956*** (0.204)		
Climate-relevant bureaucrat	44.587*** (9.421)	44.112*** (9.399)	73.150** (25.560)	62.079** (19.771)
N	2,276	2,276	9,379	6,591
Adjusted R ²	0.884	0.884	0.511	0.735
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
Agency FE	✓	✓		
GS-agency FE			✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 1: Regressions of the proportion of staff with a graduate degree on firm demand in the preceding quarter (logged count of job postings), interacted with the climate relevance of job type. Models (1–2) pool across GS grades within agencies; models (3) and (4) separate staff within agencies by GS grade, limited to mid-level grades (GS-8 through GS-12) or upper-level grades (GS-13 and above). All models control for the interaction of climate relevance with software demand. Standard errors clustered by agency.

Results reported in Table 1 align with our theory. Pooling across all GS grades, we find that firm demand for climate workers reduces the share of climate-relevant bureaucrats with graduate degree relative to other bureaucrats in the same agency. The effect is of non-trivial magnitude. Model 1 indicates that the change in firm demand observed between 2020Q4 and 2024Q4 would be associated with a 1.4 percentage point reduction in the share of bureaucrats with graduate degrees relative to others in their agencies, an 18% narrowing of the standard education gap between climate-relevant and other bureaucrats.³⁴ Similar results are found when focusing on firm demand

³³We use Lightcast’s coding of jobs that require skills in software development or engineering.

³⁴Across the sample, climate-relevant bureaucrats are eight percentage points more likely to hold a graduate degree than other bureaucrats.

for the specific types of climate workers recently sought by the federal government (Model 2).³⁵

Models 3–4 include GS grade-by-agency fixed effects, thus comparing climate-relevant bureaucrats to others in their agency and of the same professional rank. Model 3 is limited to mid-level workers (GS-8 through GS-12) and Model 4 to upper-level workers (GS-13 and above). Results indicate that firm demand reduces relative educational attainment most clearly among upper-level bureaucrats. Among upper-level workers, the change in firm demand between 2020Q4 and 2024Q4 would be associated with a 2.9 percentage point reduction in the relative share with graduate degrees. The estimate for mid-level workers is substantially smaller and statistically insignificant. The concentrated effect on upper-level workers is consistent with our argument: the highest quality climate experts — those commanding the largest salaries — are those whom the government has the hardest time acquiring and retaining amid rising firm demand.

Retention and Recruitment

The results in Table 1 suggest that private sector demand for climate expertise produces a relatively lower quality climate bureaucracy. There are two primary paths through which private sector demand could shape the federal workforce. First, private sector demand may lead high-quality experts to exit government in pursuit of greater private sector pay (failed *retention*). Second, more abundant and lucrative private sector opportunities could make it more difficult for the government to *recruit* climate talent.

To determine how much of the aggregate effect of private sector demand is driven by challenges in retention versus recruitment, we make use of additional FedScope datasets on (a) bureaucrats who resigned their positions and left the federal workforce and (b) newly hired federal workers. We estimate models of the same structure as those in Table 1, with and without GS grade-agency

³⁵We find consistent results when respecifying the outcome as the count of graduate degree-holding bureaucrats (Table G1). We prioritize the proportional outcome to characterize the quality of the average bureaucrat, per the theory. We also find consistent results without quarter-year fixed effects and with a control for presidential transitions (Table G2).

fixed effects, redefining the outcome as logged counts of quits and hires. We additionally estimate quits and hires for each GS grade individually.

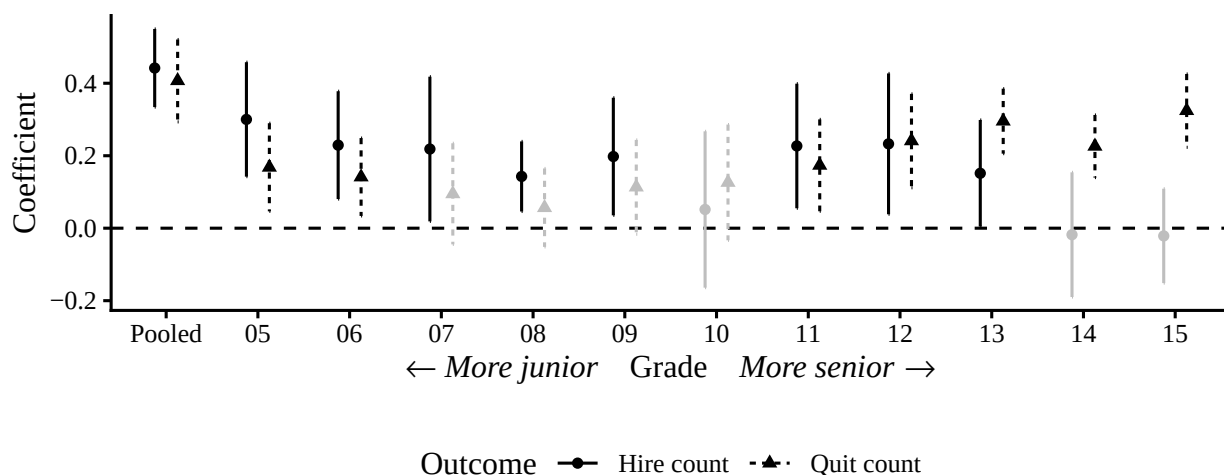


Figure 3: Effect of firm demand for climate expertise on hires and quits of climate-relevant bureaucrats relative to other bureaucrats, by GS grade. Difference-in-differences estimates; each coefficient estimated in separate model. Results significant at $p < 0.05$ darkened. Higher values indicate that firm demand increases hires/quits of climate-relevant bureaucrats relative to other bureaucrats. Full results in Appendix Table H1.

Results indicate that firm demand increases quits of climate-relevant bureaucrats and, consistent with efforts to compensate for those quits, hires of such bureaucrats (Appendix Table H1).³⁶ However, as illustrated in Figure 3, there is variation across GS grades in the extent to which recruitment keeps pace with resignations. Firm demand increases quits and hires to similar extents across lower and middle ranks. At more senior ranks, however, firm demand increases quits more so than hires; at the two senior-most GS grades, firm demand has no significant effect on hires. This suggests challenges in recruiting new upper-level climate bureaucrats, those attracting the largest private sector salaries, to make up for resignations induced by private sector demand. This provides further evidence that firms drive a net outflow of experts from upper bureaucratic ranks.³⁷

³⁶Personnel turnover itself pressures bureaucracies, insofar as expertise grows with experience.

³⁷Can this personnel turnover explain the reductions in average educational attainment reported in Table 1? Additional tests suggest so. At higher ranks, firm demand increases resignations among staff with graduate degrees, not among less educated staff. Firm demand does not, however,

Destinations of Climate Experts

Are outflows of climate expertise from the government more sensitive to demand from firms with strong climate records (those that received an A grade from CDP)? Under the assumption that experts work on more pro-social goals at top climate performers, the answer to this question is relevant for assessing the welfare effects of the labor market competition that we document.

We re-estimate the model of quits reported in Figure 3 after separating job postings by A-grade and sub-A grade firms. Two conclusions can be drawn from the results, reported in Appendix I.1. First, across ranks, demand from A-grade firms increases quits of climate-relevant bureaucrats relative to colleagues of the same rank and within the same agency. Second, independent of demand from A-grade firms, demand from firms with *sub-A* climate marks likewise produces relatively more quits among climate-relevant government staff. Demand from such firms increased by 112% between 2020Q and 2024Q; this would predict 28% more quits among upper-level climate-relevant bureaucrats and 16% more quits among mid-level climate-relevant bureaucrats. We do not find that demand from sub-A grade firms significantly increases quits among lower-level staff.

These results suggest that there is a meaningful flow of experts from government to firms with lagging records on climate. While the marginal effect of A-grade job postings may exceed that of sub-A grade postings (Figure I1), level differences in demand across these types of firms point to substantial flows of expertise from government towards climate laggards.³⁸

Alternative Explanation

There may be factors varying over time and correlated with private-sector demand that affect climate-relevant bureaucrats differently than their non-climate colleagues. Quarter-year fixed ef-

increase *hires* of workers with graduate degrees at any GS grade, indicating that firm demand produces a net outflow of more educated staff at senior ranks (Appendix Tables H2–H3).

³⁸One possibility is that laggards seek experts to upgrade their climate practices, which would attenuate the societal welfare loss of this expertise reallocation. We do not find evidence that climate hiring predicts improvements in CDP grades, however (Table I3).

fects, which we use in all models, should absorb time-variant nationwide confounders. Identification relies on the absence of residual within-quarter confounders. We address below what we consider to be the two most plausible confounding trends: broad structural shifts in the U.S. labor market for high-skilled service workers, and shifts in presidential prioritization of climate.

The first possibility we address via inclusion of the software placebo interaction control in each model, which allows us to distinguish climate-specific labor market competition from general growth in demand for technically skilled workers. The second possibility is that our results conflate the effect of firm demand with shifts in presidential prioritization of climate, given the spike in firm demand around the 2021 Trump-Biden transition and Biden’s own climate policy initiatives. It is likely that such policy changes affected firm hiring priorities; the Biden administration may have also expanded the ranks of junior staff, thus reducing average education levels across the climate bureaucracy in the aggregate. But this explanation is inconsistent with the reduced bureaucratic quality that we identify *within* GS grades (Table 1); it is unclear why pro-climate administrations would pursue less educated staff to fill upper-level civil service positions. Moreover, this explanation is inconsistent with the increased quit counts, and quit counts in excess of new hires at upper-level GS grades, depicted in Figure 3. Politically motivated exits of climate-relevant bureaucrats are most plausible under climate-skeptical populist governments, not pro-climate administrations (Carnegie, Clark, and Zucker 2024).

Readers may alternatively be concerned that efforts by the Trump administration to eject climate experts from government confound the results. Here we note that the initial spike in private sector demand occurred at the very end of the first Trump administration, after efforts to dismantle pro-climate policy work had manifest.³⁹ Our sample additionally ends prior to the second Trump administration, during which such efforts have been more pronounced.

Additional tests show that firm demand increases the extent to which environmental bureaucracies contract with private sector consultants. As detailed in Appendix J, this growth is not consistent with the changes in contracting that stem from shifts in presidential priorities. It is,

³⁹*Union of Concerned Scientists* 2021 [perma.cc/46NV-63F2].

however, consistent with an effort by government agencies to substitute for diminished in-house expertise with consultants. Given work that links reliance on consultants to reductions in bureaucratic capacity and the privatization of policy knowledge (Keele 2019), we also see such results as further evidence that firm demand erodes the quality of the climate bureaucracy. This increased use of consultants also suggests that informational lobbying alone cannot compensate for the public-private expertise gap.

QUALITY OF CLIMATE GOVERNANCE

We have thus far demonstrated that firm demand for climate expertise constrains the government's access to high-quality in-house experts. We now consider whether that demand diminishes the ultimate quality of climate governance. To measure quality, we consider two dimensions of climate governance: government production of climate information, proxied via citations of government climate research, and enforcement of climate regulations.

Government Research

To measure the quality of information produced by the government, we use counts of citations to individual research works, a common metric of research productivity (e.g., Barceló et al. 2025). We obtain these data from OpenAlex, which provides metadata on hundreds of millions of scientific works, including citation counts, topics, and author affiliations. OpenAlex is among the most comprehensive bibliographic catalogs; unlike Google Scholar, it is open source and makes its full database available for download, allowing for reproducible data collection (Walters 2025). We limit the sample to work published between 2010–2024 with at least one U.S.-based author. With these metadata, we test whether firm demand reduces citations of U.S. government-authored climate works relative to U.S. government-authored *non*-climate works. We measure climate relevance via OpenAlex topic tags.⁴⁰ In a second test, we examine whether firm demand reduces

⁴⁰We use all topic tags that concern climate: climate change policy and economics (T10471); species distribution and climate change (T10895); climate change communication and perception (T11488); climate change, adaptation, migration (T12656); flooding and environmental impact

citations of government climate research relative to non-government climate research.

We identify 6,721 works of climate research authored by at least one U.S. government researcher. In the first test, we contrast these with a random sample of 44,266 non-climate governmental works.⁴¹ We estimate models of the following form by OLS:

$$\text{Citations}_{ikt} = \beta [\text{demand}_{t-1} \times \text{climate relevance}_i] + \gamma \mathbf{Z}_{it} + \delta_{kt} + \varepsilon_{ikt}$$

This model regresses the log-transformed stock of citations to work i , within field k and published in quarter-year t , on the climate relevance of that work interacted with firm demand a year prior to its publication date. We control in the vector $\mathbf{Z}_{it}$ for the constituent terms of the interaction, as well as the interaction of climate relevance with the software demand placebo, and include a field-quarter-year fixed effects term δ_{kt} (fields being, for example, chemistry or economics). Note that all citation stocks are measured as of March 2026; we include field-quarter-year fixed effects, thus comparing works published at the same time and in the same field, to account for field-specific differences in patterns of citation accumulation over time.

The second test instead compares citations of government climate research to citations of non-government climate research (the latter totaling nearly 70,000 works). Non-government research is defined as work with no U.S. government author. In this test, we estimate the following model:

$$\text{Citations}_{ikt} = \beta [\text{demand}_{t-1} \times \text{government authorship}_i] + \gamma \mathbf{Z}_{it} + \delta_{kt} + \varepsilon_{ikt}$$

As before, the outcome is the log-transformed citation stock for work i . We again include the software placebo interaction control, constituent interaction terms, and field-quarter-year fixed effects.

Table 2 reports results in line with expectations. Results of Model 1 indicate that as firm demand for climate expertise increases, climate research subsequently published by government authors accrues fewer citations relative to government research on other topics published at the

(T13969). Research works defined as articles, books and book chapters, preprints, and reviews.

⁴¹We randomly sample 1,200 works per quarter with at least one government author, then filter to works where that government author is based in the U.S., yielding the final sample of 44,266.

	Citations (log)	
	(1)	(2)
Demand $\times$ climate relevance	-0.162* (0.068)	
Demand $\times$ government authorship		-0.107** (0.039)
Climate relevance	0.344 (0.804)	
Government authorship		0.657 (0.614)
Sample	Government research	Climate research
N	47,321	73,367
Adjusted R ²	0.150	0.124
Software demand control	✓	✓
Quarter-year-field FE	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 2: Regressions of log citation counts on the interaction of private sector demand and (a) the climate relevance of a given work, and (b) U.S. government authorship of that work. Fixed effects specification reflects quarter-year of publication and disciplinary field of a given work. Standard errors clustered by quarter-year of publication.

same time. These results imply that doubling firm demand would reduce total citations to government climate research by 11% relative to non-climate research. Results of Model 2 likewise indicate that as firm demand grows, citations of government climate research decline in comparison to non-government climate research published in the same fields and at the same time. In this case, doubling firm demand would predict a 7% relative reduction in government citation stocks. While these results do not point to absolute reductions in citations to government climate research, they do indicate both a rebalancing of government research away from climate and a shift in the locus of climate research away from government.⁴²

⁴²Results are robust to lagging private sector demand by two and five years. We additionally find that the share of upper-level bureaucrats with a graduate degree positively predicts citations to subsequently published work by a given agency (Appendix K).

Regulatory Enforcement

We lastly consider the quality of regulatory enforcement. To do this, we focus on rules issued via two federal laws that regulate industrial pollution: the 1963 Clean Air Act (CAA) and 1972 Clean Water Act (CWA). Enforcement of the CAA and CWA is overseen by the Environmental Protection Agency (EPA), often via its regional field offices in coordination with state-level agencies.⁴³

The following analyses ask whether firm demand predicts growth in the number of facilities with “unresolved violations” of the CAA relative to the number with unresolved violations of the CWA. Unresolved violations refer to instances where facilities are out of compliance with a regulation, but where enforcement actions, such as the imposition of a fine, have not concluded; i.e., cases where neither “the facility [nor] the regulatory authority has ... performed an action to correct a violation.”⁴⁴ We see such unresolved violations as indicative of constrained bureaucratic capacity within enforcement agencies; supporting this interpretation, the EPA’s inspector general has linked decline in EPA enforcement actions to losses of enforcement staff.⁴⁵ We conduct these analyses at the state-quarter-year level, including measuring firm demand by state, to match the state-level nature of CAA and CWA enforcement.⁴⁶

Comparing the CAA to the CWA allows us to account for within-state, time-variant confounders of the link between firm demand and enforcement quality, such as local regulators’ gen-

⁴³EPA 2023 [perma.cc/L3ZZ-3WAD]. Enforcement authority is largely delegated to state agencies by the EPA (EPA 2026, perma.cc/4N65-LKZS)

⁴⁴EPA [perma.cc/W9JN-WZ59; perma.cc/GG3K-U3AA]. We gather data on unresolved violations from the EPA ECHO database. Violations are coded as unresolved at quarter-year t if they were detected at or prior to t , with no resolution as of t . We focus on violations reported to the EPA, which include violations of emissions and effluent limits and procedural violations, such as failure to install required monitoring equipment (EPA 2014, perma.cc/MQR8-5EP3).

⁴⁵EPA *Office of the Inspector General* 2020 [perma.cc/5XW6-XL63].

⁴⁶Conducting tests at the state-quarter-year level also increases statistical power, as demand for climate expertise varies across states within the same quarter.

eral interest in environmental protection.⁴⁷ Enabling this comparative approach is the fact that the CAA, not the CWA, is a primary vehicle for climate-related rule-making. The CWA, while reflecting a general interest in environmental protection, focuses on issues of local water pollution and not on global climate change or greenhouse gas emissions. Accordingly, we see the CAA as more likely to be affected by climate hiring than the CWA.⁴⁸

We estimate the following model by OLS:

$$\text{Unresolved violations}_{lst} = \beta [\text{CAA}_l \times \text{firm demand}_{st}] + \zeta \mathbf{Z}_{lst} + \gamma_s + \delta_t + \varepsilon_{lst}$$

The model regresses the log count of facilities with unresolved violations of law l (CAA or CWA), in state s in quarter-year t , on the interaction of a binary indicator of the CAA with state-level firm demand for climate experts lagged by one quarter. We control for the constituent interaction terms, as well as the interaction of the CAA binary with software demand. Given the possibility that growth in unresolved violations is an artifact of growth in violations, rather than decline in bureaucratic capacity, we control for the log count of new violations by state-quarter-year, interacted with an indicator for violations of the CAA. We include state and quarter-year fixed effects.

Results in Table 3 accord with expectations: firm demand limits the enforcement of climate-related regulations. As firm demand for climate experts increases within states, the number of facilities with unresolved CAA (i.e., climate-relevant) violations increases relative to the number

⁴⁷Pro-environment governments may spur firm hiring of environmental specialists, of which climate experts are a type, and simultaneously invest more in CAA and CWA enforcement. Alternatively, shareholders or consumers may push firms to alter environmental performance, which may affect both hiring and regulatory compliance.

⁴⁸NRDC 2026 [perma.cc/8HEN-ZZM4]. This is seen in EPA descriptions of CAA and CWA compliance monitoring. CAA monitoring encompasses direct and indirect greenhouse gases, such as chlorofluorocarbons and ozone (EPA 2026, perma.cc/BUP5-U5Z2). CWA monitoring largely focuses on wastewater management, dredging, and spills of hazardous substances, issues that do not require climate-related skills (EPA 2026, perma.cc/K9ZG-2PHV).

	Facilities with Unresolved Violations (log)	
	(1)	(2)
Demand $\times$ CAA	0.423* (0.159)	0.279** (0.103)
Firm demand (ln)	-0.277* (0.129)	-0.177+ (0.089)
CAA = 1	-13.399*** (3.442)	-10.499*** (2.587)
N	5,220	5,220
Adjusted R ²	0.541	0.696
Software demand control	✓	✓
New violations control		✓
State FE	✓	✓
Quarter-year FE	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: Regressions of log count of facilities with unresolved violations in a given state-quarter-year on the interaction of firm demand and a binary indicator violations of the Clean Air Act (CAA) versus the Clean Water Act. Standard errors clustered by state.

with unresolved violations of the CWA. Point estimates for Model 1 indicate that doubling state-level firm demand would increase the count of facilities with unresolved CAA violations by 34% relative to the CWA. These results are robust to controlling for the count of new violations in a given state (Model 2), as well as when iteratively excluding each state from the sample (Appendix Figure L1). The results are not a byproduct of intentional underenforcement of the CAA by the Trump administration; the Trump administration also oversaw cutbacks to CWA enforcement (the comparison group), and quarter-year fixed effects should absorb any such nationwide shocks.⁴⁹

CONCLUSION

Growth in the climate regime complex is amplifying global demand for climate expertise. But the supply of that expertise is limited. This paper reframes climate governance as the product of a labor market competition between governments and firms for climate experts. We argue and find that as firm demand increases and private sector wage premia mount, governments struggle in the short term to recruit and retain high-quality experts. Firm demand in turn impairs public climate

⁴⁹NPR 2020 [perma.cc/NY2R-77F4].

governance, measured via the impact of government climate research and extent of climate-related regulatory enforcement.

While we test this theory in the U.S., we see it as broadly applicable. First, in a number of high-income countries, there exist meaningful private sector wage premia for the highest skilled workers, as well as ascendant private sector demand for climate expertise.⁵⁰ Second, the labor market competition we characterize is transnational — a significant share of firms hiring climate experts on the U.S. market are headquartered elsewhere. Corporate reforms in one country may thus have spillover effects on bureaucratic capacity in another. Third, international organizations such as the IMF have, like national governments, sought to hire greater numbers of climate experts in recent years (Committeri et al. 2022). Growth in private sector demand across high-income countries plausibly complicates efforts by these institutions to bolster bureaucratic expertise.

This paper points to three paths for future work. First, for tractability, our theory treats changes in firm demand for expertise as exogenous. Future work should problematize this. One possibility is that firm demand is endogenous to climate policy trajectories. When firms experience or anticipate stronger climate regulations, the returns to acquiring expertise plausibly grow, increasing wage offers and accelerating the flow of experts away from government. Policy ambition and regulatory stringency may, as a result, paradoxically complicate the implementation of those laws.

Second, we theorize about and measure short-term changes in bureaucratic expertise, arising from an initial shortage of climate experts. Over time, mounting demand for expertise should increase the supply of that expertise, shifting labor market power away from expert workers. This may incentivize experts to distinguish themselves to employers in some way. For those interested in private sector work, one means of doing so may be to undertake government work — acquiring political connections and bureaucratic knowledge that subsequently appeal to private employers — thus setting a revolving door between the public and private sectors into motion. While climate labor markets have not yet reached that stage in practice, scholars should remain attentive to the evolution of climate as a profession and its implications for the efficacy of climate governance.

⁵⁰*LinkedIn* 2024 [perma.cc/B2QT-DM2K].

Third, while we develop and test the argument in reference to climate, we see the logic as applicable to other issues that are technically complex, which creates demand for experts across sectors, and novel as hiring priorities, implying a scarce supply of such experts. Issues such as artificial intelligence likely satisfy these conditions. As firms mobilize in novel and complex policy domains, states may struggle to access the bureaucratic expertise needed to govern well.

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APPENDICES

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A. FORMALIZATION OF THEORY

Here we develop a simple formal model of a labor market in which the government G and two firms (FA, FB) compete to hire a climate expert E . The expert is endowed with some exogenous level of quality, $q > 0$, which is observable to all potential employers. We assume that all potential employers want to hire high-quality experts. Employers' utility for hiring the expert is increasing in the demand parameter, $d > 0$, which reflects exogenous market-wide demand for climate expertise.

To compete for E , the potential employers begin the game by making wage offers ω_i ($i \in \{G, FA, FB\}$). E then selects one of the three offers to accept, and payoffs accrue. Each of the potential employers receives $dq - \omega_i$ if E accepts their offer, and 0 otherwise. If E accepts either FA or FB 's offer, her utility is simply the wage ω_{FA} or ω_{FB} . However, E is also endowed with some level of pro-sociality $\gamma > 0$; this can be understood as the non-wage, intrinsic premium placed on doing meaningful, welfare-enhancing work on climate, or the strength of their public service motivation. We assume that highly pro-social individuals are willing to trade off higher pay for the opportunity to work at G , where they would serve the public interest rather than the interests of corporate shareholders.¹ E receives a payoff of $\omega_G + \gamma$ if she accepts G 's offer.

¹We do not claim that private sector climate efforts do not serve the public interest in any way. We merely note that firms' primary responsibility is to serve shareholders rather than the public. An

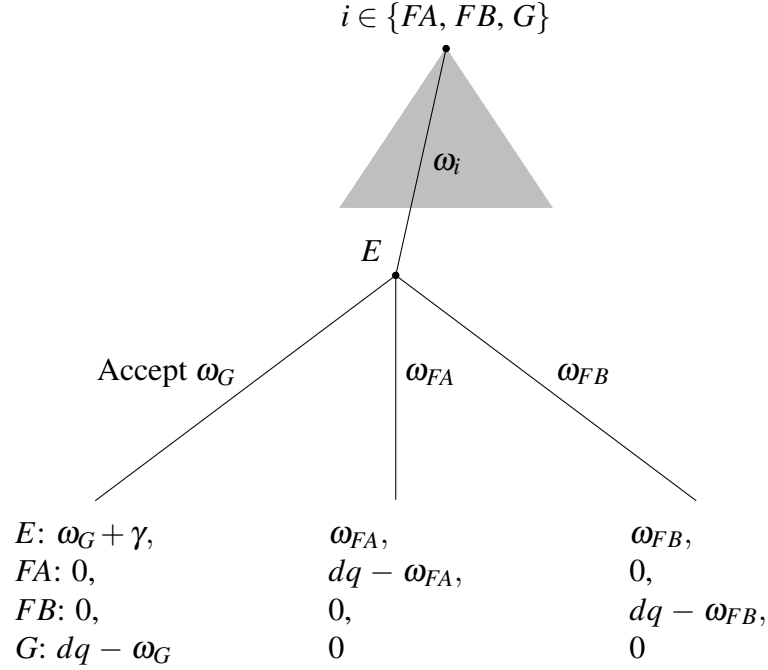


Figure A4: A model of public-private competition for climate experts.

We begin by characterizing the distinguished subgame perfect Nash equilibrium for this game when there are no constraints on the government's wage offers.² Beginning in the second stage, E accepts the max value of the set $\{\omega_G + \gamma, \omega_{FA}, \omega_{FB}\}$. Thus, in order to make E indifferent between its offer and that of the best private sector offer, ω_F , G only needs to offer $\omega_G = \omega_F - \gamma$. The highest wage that the identical firms would receive positive utility from paying is $\omega_F = dq - \varepsilon$. Substituting this value of ω_F into the government's wage offer, we can identify G 's equilibrium wage offer (taking the limit as $\varepsilon \rightarrow 0$): $\omega_G^* = dq - \gamma$. In this unconstrained equilibrium, G can hire experts while paying below-private sector rates due to the non-wage value of public employment.

Next, we consider the more realistic scenario in which the government faces an upper bound, $\bar{\omega}_G$, on the wage offers it can make to E . Research points to a consistent private sector wage

interesting extension of the model could treat γ as a function of both individual expert preferences as well as the pro-sociality of particular governments and firms, to allow for the fact that some governments and firms are more pro-social than others.

²There are multiple equilibria in auction models of this sort. By “distinguished,” we refer to the equilibrium where FA and FB bid their valuation of E , $\omega_F = dq - \varepsilon$.

premium for the most educated workers, with government employers reacting sluggishly to private sector wage growth; further, government wage offers are often constrained by official salary schedules that determine the maximum compensation for a given position or career grade.

The public sector wage constraint $\bar{\omega}_G$ is not binding when it is greater than G 's equilibrium offer in the unconstrained game. However, when $\bar{\omega}_G < \omega_G^*$, G can no longer compete with the private sector even after factoring in the pro-social bonus of government employment. Since there is still competition within the private sector when G cannot make a competitive offer, both firms continue to offer $\omega_{FA} = \omega_{FB} = dq - \varepsilon$ and E goes to the private sector. Formally,

Proposition 1. *When $\bar{\omega}_G < dq - \gamma$, there exists no equilibrium in which E accepts G 's offer.*

Having established this condition, we note that — for any $\bar{\omega}_G$ — G 's ability to compete with the private sector to hire climate experts is declining in the term $dq - \gamma$. From here, it is straightforward to make predictions regarding the likely effects of increased demand for climate expertise on the characteristics of the government's climate bureaucracy. First, since $dq - \gamma$ is increasing in d , increased demand for climate expertise should increase the probability that experts choose private over public employment.

Corollary 1. *Since $\frac{\partial dq - \gamma}{\partial d} > 0$, higher demand for climate expertise makes experts more likely to go to the private sector.*

Second, we observe that the relationship between $dq - \gamma$ and d is affected by expert quality q . Higher levels of q amplify the effect of d on offered wages; increased demand for climate expertise has the largest effect on the market wage for high-quality experts. Thus, as demand increases, higher quality experts become relatively more likely to go to the private sector.

Corollary 2 (Hypothesis 1). *Since $\frac{\partial^2(dq - \gamma)}{\partial d \partial q} > 0$, higher demand for climate expertise makes higher quality experts relatively more likely to go to the private sector.*

Third, we examine the relationship between demand for climate expertise, pro-sociality γ , and the probability that E goes to the private sector. By rearranging terms of the inequality in Proposition 1, it can be shown that G can make competitive offers when $\gamma \geq dq - \bar{\omega}_G$. Intuitively, G can

compete for climate experts when they are sufficiently pro-social to make up for the gap between the private sector wage dq and the maximum public sector wage $\bar{\omega}_G$. As demand rises, the level of pro-sociality necessary for G to remain competitive increases as well. Thus, increased demand (and accordingly the private sector wage available to experts) is likely to “screen out” less pro-social experts from government, increasing the average pro-sociality of the climate bureaucracy.

Corollary 3 (Hypothesis 2). *Since $\frac{\partial dq - \bar{\omega}_G}{\partial d} > 0$, higher demand for climate expertise should increase the average pro-sociality of government climate experts.*

We now consider an extended version of the model, accounting for the possibility of government contracting with the private sector. In this extension, if E accepts a firm’s offer and goes to the private sector, G can contract with E ’s employer to procure E ’s consulting services. If G chooses to do so, it receives the payoff $dq - \omega_F$, the same payoff as the firm that hired E ; if it chooses not to do so it receives 0. Crucially, the upper bound on G ’s wage offer does not apply to contracting; G offers ω_F at this stage even if $\omega_F > \bar{\omega}_G$.

Contracting is not an attractive option for G when it has the option to hire an expert directly (i.e., when $\bar{\omega}_G \geq dq - \gamma$), because contracting would lead G to forfeit its pro-sociality subsidy γ by paying the higher private sector wage. However, when $\bar{\omega}_G < dq - \gamma$ and the government cannot hire the expert in-house, contracting becomes a favorable option. Again, it is clear from this expression that this is more likely to be the case as demand increases.

Corollary 4 (Hypothesis 3). *Increased demand for climate expertise increases the government’s procurement of climate-related consulting services from the private sector.*

B. PROOFS

Unconstrained game The game we analyze is a modified first-price sealed-bid auction with complete information: potential employers submit “bids” to an expert, knowing each other’s valuations of the expert’s labor, and the employer who submits the highest bid wins the expert’s labor at the price bid. In the unconstrained game, we add two modifications to the standard first-price

sealed-bid auction: we model the expert who receives the wage bids as a (weakly) strategic actor, and we introduce a parameter γ that increases the expert's utility for accepting G 's wage offer.

To facilitate the comparison to the first-price sealed-bid auction, we introduce the concept of *total bid value* (U_E^i): this is the total utility that E receives from the wage and nonwage benefits of potential employer i 's offer, and the basis upon which E chooses among its various offers. For each firm, FA and FB , total bid value is equivalent to the wage offer ω because their offers have no nonwage component ($U_E^{FA} = \omega_{FA}$, $U_E^{FB} = \omega_{FB}$). For G , total bid value is equal to the government's wage offer plus the nonwage pro-sociality benefit of government employment: $U_E^G = \omega_G + \gamma$.

Having introduced total bid value, we can now consider different potential employers' *valuations* of E 's labor. An employer's valuation for hiring an expert E is simply the positive part of their payoff if E is hired. Each firm receives dq if it hires E . Since neither q nor d varies between firms, both firms value E 's labor at dq ($v_{FA} = v_{FB} = dq$). G also receives dq if it hires E ; however, when G offers ω_G , E receives $\omega_G + \gamma$ if she accepts. Thus, γ functions as a sort of subsidy (or, alternatively, a handicap) for governmental hiring.

We can move γ into G 's utility function by noting the following. Thanks to γ 's subsidizing effect, G does not have to set $\omega_G = U_E^G$ in order to ensure that its total bid value is equal to U_E^G . Rather, to ensure that E receives a given value of U_E^G , G must only offer $\omega_G = U_E^G - \gamma$. We can then express G 's utility for making any wage offer with total bid value of U_E^G , if that offer is accepted, as follows: $dq - (U_E^G - \gamma) = dq + \gamma - U_E^G$.

This makes clear that, in the unconstrained game, G values E 's labor at $v_G = dq + \gamma$. This is because, as γ functions as a substitute for wage compensation, G is able to capture this value by reducing its wage offers by γ . Further, since we have assumed that $\gamma > 0$, the potential employers' valuations satisfy the following: $v_G > v_{FA} = v_{FB}$.

It is clear that the potential employer with the highest valuation will make the offer with highest total bid value, and in turn hire the expert, in any equilibrium. To see why, consider an outcome in which G makes an offer that is lower than one of the firm's offers, which in turn is less than or equal to the firm's valuation ($\omega_G + \gamma < \omega_F \leq dq$). In this case, G could profit by making any

offer ω_G such that $\omega_F \leq \omega_G + \gamma \leq dq + \gamma$, which would guarantee G positive utility as opposed to 0 (the payoff for failing to hire E). Likewise, consider an outcome in which both G and a firm make offers that are greater than the firm's valuation, and in which the firm's offer is higher than the government's ($dq < \omega_G + \gamma < \omega_F$). In this case, the firm hires E but receives negative utility; it would therefore prefer to lower its offer to $\omega_F \leq dq$ and receive 0, resulting in G hiring E .³

We characterize what Osborne (2004) calls the “distinguished” equilibrium of this game as follows. First, assume that when E receives n offers with identical utility, she chooses each of them with probability $\frac{1}{n}$. Both of the identical firms make the highest wage offers that, if accepted, would provide them positive utility: $\omega_{FA}^* = \omega_{FB}^* = dq - \varepsilon$. G would find it profitable to make any offer such that $dq - \varepsilon < \omega_G + \gamma < dq + \gamma$. G will make the lowest possible offer with a total bid value greater than $dq - \varepsilon$; this offer is $\omega_G^* = dq - \gamma$, resulting in a total bid value of dq for E . In this equilibrium, E gets a payoff of $(dq - \gamma) + \gamma = dq$; G hires E and receives the payoff $dq - (dq - \gamma) = \gamma$; both firms get a payoff of 0. ■

Proposition 1 We now consider a version of the model in which G faces an upper bound ($\bar{\omega}_G$) on the wage that it can offer to any expert. Intuitively, the government wage ceiling is only binding when it is lower than the wage that G would have offered absent the constraint (e.g., when $\bar{\omega}_G < dq - \gamma$). When this is not the case, such that $\bar{\omega}_G \geq dq - \gamma$, the game and its equilibria are identical to those of the unconstrained version.

Proposition 1 states, however, that G cannot hire E in equilibrium when the wage constraint binds ($\bar{\omega}_G < dq - \gamma$).⁴ The proof is simple: when $\bar{\omega}_G < dq - \gamma$, the maximum total bid value that G can offer to E is $U_E^G = \bar{\omega}_G + \gamma$. However, when $\bar{\omega}_G < dq - \gamma$, it is clear that $\bar{\omega}_G + \gamma < dq$. Assuming that $dq - \bar{\omega}_G - \gamma > \varepsilon$, G is unable to make an offer that E prefers to the firms' maximum profitable offer of $\omega_{FA} = \omega_{FB} = dq - \varepsilon$.

Further, because both firms have identical valuations, both firms will make the maximum profitable offer of $\omega_{FA} = \omega_{FB} = dq - \varepsilon$ in all equilibria. If either firm makes a wage offer greater than

³This discussion is adapted from Osborne 2004 (Oxford University Press).

⁴More completely, when $\bar{\omega}_G < dq - \gamma - \varepsilon$.

$dq - \varepsilon$, its utility can only either be 0 (if $\omega_F = dq$) or negative (if $\omega_F > dq$) and it would thus prefer to lower the offer to $dq - \varepsilon$. If one firm makes an offer lower than $dq - \varepsilon$ while the other one makes the offer $dq - \varepsilon$, the firm with the lower offer could raise its utility from 0 to $\frac{\varepsilon}{2} > 0$ by raising its offer to $dq - \varepsilon$ (recall that we are assuming ties are broken with equal probability).

Thus, when $\bar{\omega}_G < dq - \gamma$, the government's offer is irrelevant and the expert's wage is decided by private sector competition. Both firms will offer $\omega_{FA} = \omega_{FB} = dq - \varepsilon$; G offers any $\omega_G \leq \bar{\omega}_G$; E accepts both firm's offers with equal probability. E receives utility of $dq - \varepsilon$, both firms receive $\frac{\varepsilon}{2}$ in expectation, and G receives 0. ■

Corollary 1 Proposition 1 states that G cannot hire climate experts when $\bar{\omega}_G < dq - \gamma$. Corollary 1 claims that, all else equal, this inequality is more likely to be satisfied as demand for climate expertise increases. To show this, we can take the partial derivative of $dq - \gamma$ with respect to d :

$$\frac{\partial dq - \gamma}{\partial d} = q \quad (1)$$

Since $q > 0$, the inequality $\bar{\omega}_G < dq - \gamma$ is more likely to hold as d grows. Thus, as demand for climate expertise increases, climate experts become less likely to work for the government. ■

Corollary 2 Corollary 2 claims that, all else equal, higher quality experts will be relatively more likely to go to the private sector following increases in demand for climate expertise. To see why, we can begin as in Corollary 1 by differentiating $dq - \gamma$ with respect to d :

$$f_d = \frac{\partial dq - \gamma}{\partial d} = q \quad (2)$$

We can then take the cross-partial of f_d with respect to q :

$$f_{dq} = \frac{\partial f_d}{\partial q} = q^{1-1} = 1 \quad (3)$$

Since $1 > 0$, we can see that the effect of an increase in d on the term $dq - \gamma$ is increasing in q . ■

Corollary 3 Corollary 3 claims that, all else equal, an increase in demand for climate expertise will increase the level of pro-sociality necessary for experts to accept government employment. To prove this, we begin by inverting the inequality in Prop. 1; experts *will* opt for government

employment when:

$$\bar{\omega}_G \geq dq - \gamma \quad (4)$$

Expressed in terms of γ , we have $\gamma \geq dq - \bar{\omega}_G$. As the right-hand side of this inequality grows larger, the value of γ necessary to keep the inequality satisfied must grow as well. To determine how an increase in d will affect the size of $dq - \bar{\omega}_G$, we can take the partial derivative of this expression with respect to d :

$$\frac{\partial dq - \bar{\omega}_G}{\partial d} = q \quad (5)$$

Since $q > 0$ by definition, an increase in d (holding q and $\bar{\omega}_G$ constant) will necessarily increase the value of γ necessary for E to accept G 's offer. ■

C. IDENTIFICATION OF CLIMATE-RELATED JOBS

Our primary approach to identifying climate-related postings is based on a keyword search of job descriptions. We search for the following set of terms (note that all included a wildcard suffix to ensure that the search captured both, e.g., “air pollution” and “air pollutant”): air pollut; air quality; atmospheric; climate action; climate adapt; climate analysis; climate change; climate change adapt; climate change mitig; climate data; climate engineer; climate information; climate mitig; climate model; climate polic; climate predict; climate resilien; climate research; climate system; climate variab; climatolog; global warming; meteorolog; net zero; weather forecast; weather model.

We validate this dictionary by comparing it to climate-related skills in the Lightcast Open Skills Taxonomy, which we identify via keyword searches (taxonomy version 9.19, dated June 28, 2024; perma.cc/4HYD-ZB5E): carbon (carbon footprint reduction, carbon accounting, carbon management, carbon markets, carbon offsets, carbon capture and storage, low carbon solutions, low carbon development), climate (climatology, climate policy, climate variability and change, climate resilience, climate modeling, climate change mitigation, climate change adaptation, climate prediction, climate engineering, climate change programs, climate information, climate analysis, Community Climate System Model, Climate Data Exchange (CDX), Action for Climate Empowerment (ACE), Climate Data Analysis Tool (CDAT)), emissions (vehicle emissions controls, emis-

sions inventory, emissions calculations, emission testing, emission standards, Continuous Emissions Monitoring Systems, fugitive emissions, emissions analyzers, emissions controls, National Emissions Standards for Hazardous Air Pollutants, stack emission measurements, emissions trading, emission reduction projects, AVOIDed Emissions And GeneRation Tool, Michigan Air Emissions Reporting Systems, UN Race-to-Zero Emissions Breakthroughs), greenhouse (greenhouse gas, Regional Greenhouse Gas Initiative), net zero (net zero).

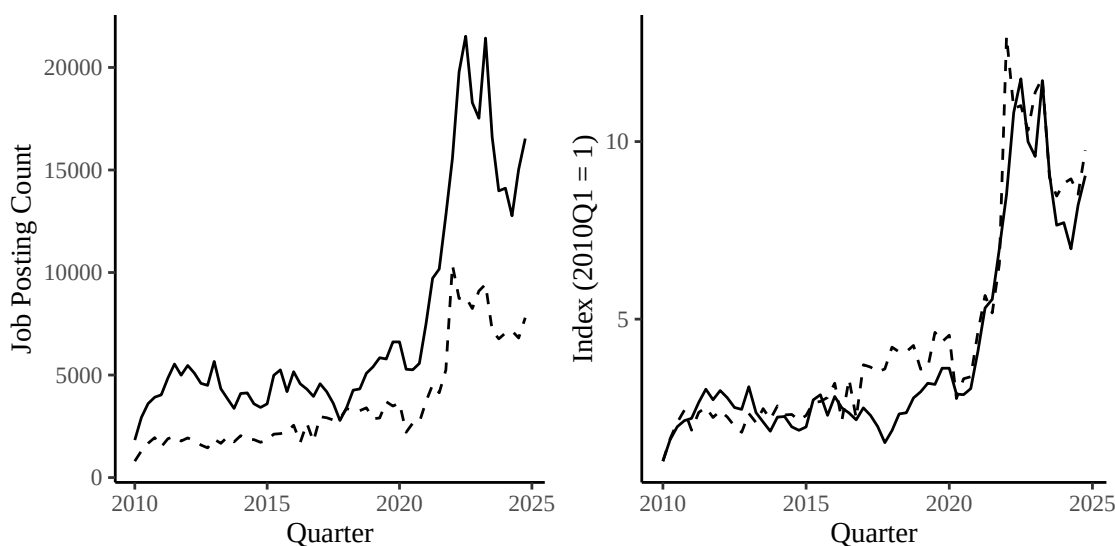


Figure C1: Counts of climate-related job postings, measured via keyword search (solid lines) and Lightcast skill coding (dashed lines).

Figure C1 shows that trends with these measures are highly correlated (correlation coefficient of 0.95). Our keyword measure, however, captures more job postings than the skill-based measure. We prefer the keyword-based measure for this reason. This measure is also more transparent; Lightcast indicates that it uses “machine-learning algorithms” and “qualitative methods” to identify skills, but does not provide further detail (see perma.cc/4HYD-ZB5E).

D. PROFILING THE FIRMS THAT HIRE CLIMATE EXPERTS

Here we assess whether climate expert-hiring firms are more likely than other firms to (a) lobby the EPA and (b) provide environmental consulting services to any federal agency. We focus on publicly traded firms for reasons of data availability (which cover 50% of climate job postings in our

data), matching firm-year level financial information from Compustat to three different databases: (a) Lightcast job posting data; (b) LobbyView federal lobbying data; and (c) data on federal procurement of environmental consulting services from the Federal Procurement Data System.

Table D1 presents the proportion of firm-years in which a firm lobbied the EPA or provided the federal government with environmental consulting services, disaggregated by whether or not they also attempted to hire climate experts in that year. First, in the aggregate, we find that firms that hire climate experts are an order of magnitude more likely than others to lobby and contract with the government on environmental issues. This is the case even *within* industries that are clearly climate-relevant; for example, across publicly traded petroleum refining firms, those that hired climate experts are twice as likely to lobby the EPA than those that did not. Similarly, while all aircraft manufacturers are exposed to climate regulations due to the carbon intensity of air travel, only those that hired climate experts provided environmental consulting services to the government. These within-industry relationships still hold after adjusting for firm size (Table D2). We do not make causal claims on the basis of these comparisons, but the results indicate that the firms that seek to hire climate experts are often the same firms that engage in the climate policy process via lobbying or consulting.

NAICS code	Industry	Proportion lobbied the EPA	
		Did not hire experts	Hired experts
22111*	Electric power generation	0.05	0.19
324110	Petroleum refining	0.11	0.20
562111	Solid waste collection	0.02	0.43
–	All industries	0.004	0.07
NAICS code	Industry	Prop. consulted gov. on environment	
		Did not hire experts	Hired experts
336411	Aircraft manufacturing	0.00	0.29
541330	Engineering services	0.01	0.43
541614	Process and logistics consulting	0.00	0.82
–	All industries	0.001	0.02

Table D1: Proportion of firm-years in which a firm (i) lobbied the EPA or (ii) signed a procurement contract to provide a federal agency with environmental consulting services, disaggregated by whether the firm attempted to hire climate experts in that year.

DV:	Federal Env. Consultant = 1			Lobbied EPA = 1		
	(1)	(2)	(3)	(4)	(5)	(6)
Postings	0.026*** (0.007)	0.026*** (0.007)	0.027*** (0.008)	0.025*** (0.003)	0.023*** (0.003)	0.026*** (0.003)
Total assets		−0.001* (0.001)	−0.001+ (0.000)		0.002*** (0.000)	0.002*** (0.001)
Employees		0.001* (0.001)	0.001 (0.001)		0.001* (0.001)	0.001 (0.001)
N	137,405	89,327	89,327	127,610	83,885	83,885
Adjusted R ²	0.188	0.199	0.281	0.125	0.132	0.125
Year FE	✓	✓		✓	✓	
Industry FE	✓	✓		✓	✓	
Industry-year FE			✓			✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table D2: Regressions of firm-year level outcomes — providing environmental consulting services for the federal government, and lobbying the EPA — on the logged number of climate expert job postings that a firm issued in a given year. Control variables are logged as well. Industry is coded at NAICS six-digit level.

E. IDENTIFICATION OF CLIMATE-RELEVANT BUREAUCRATS

We identify thirteen climate-related occupation codes, based on a review of the occupations sought in federal climate-related job postings. These occupations are environmental engineering, environmental protection assistant, environmental protection specialist, forestry, forestry technician, general natural resources management and biological science, geography, hydrologic technician, hydrology, physical science technician, physics, wildlife biology, and zoology. Not all workers in these occupations focus on explicitly climate-related tasks, though many do (*Union of Concerned Scientists* 2021, perma.cc/46NV-63F2). Lacking more precise data on individual bureaucrats’ work tasks, we see these occupations as the best metric of staff proximity to climate governance.

F. SUMMARY STATISTICS

The agencies included in the bureaucratic composition analysis dataset are the departments of the Air Force, Agriculture, Army, Commerce, Defense, Justice, Energy, Health and Human Services, Homeland Security, Housing and Urban Development, Interior, Navy, Transportation, and Veterans Affairs. Also included are USAID, the Environmental Protection Agency, General Services

Administration, National Science Foundation, NASA, and Smithsonian Institution.

Statistic	N	Mean	St. Dev.	Min	Max
Prop. MA+ (%)	2,429	50.8	17.6	18.8	100
Quit count	2,352	135	300	0	2,856
Hire count	2,352	332	841	0	10,719
Climate-relevant bureaucrat	2,429	0.5	0.5	0	1
Firm demand	2,387	6,902	5,056	1,822	21,473
Firm demand (overlapping)	2,387	3,361	2,827	452	11,645
Firm demand (software)	2,387	73,309	26,720	23,204	150,590

Table F1: Summary statistics at agency-year-job type level for analysis of bureaucratic quality, pooled across GS grades (Table 1, Figure H1). Quit, hire, and demand variables not log transformed in this table.

Statistic	N	Mean	St. Dev.	Min	Max
GS grade	37,800	8	4.32	1	15
Prop. MA+ (%)	25,114	32.8	25.6	0	100
Quit count	25,872	8.21	22.0	0	313
Hire count	25,872	19.4	57.9	0	2,856
Climate-relevant bureaucrat	37,800	0.5	0.5	0	1
Firm demand	37,170	7,139	5,125	1,822	21,473
Firm demand (overlapping)	37,170	3,422	2,797	452	11,645
Firm demand (software)	37,170	72,890	26,360	23,204	150,590

Table F2: Summary statistics at agency-year-job type-GS grade level. Quit, hire, and demand variables not log transformed in this table.

G. BUREAUCRATIC COMPOSITION: ADDITIONAL RESULTS

DV:	# Staff With Graduate Degree			
	<i>GS Pooled</i>	<i>GS: Mid Level</i>	<i>GS: Upper Level</i>	
	(1)	(2)	(3)	(4)
Demand × Climate-relevant	−2104.861* (904.647)		−150.813 (97.898)	−279.835* (103.675)
Demand (overlapping) × Climate-relevant		−1375.040* (622.537)		
Climate-relevant bureaucrat	40703.170* (16522.621)	40720.640* (16570.748)	2061.916+ (1189.740)	4269.881* (1657.587)
N	2,276	2,276	9,379	6,591
Adjusted R ²	0.598	0.598	0.589	0.623
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
Agency FE	✓	✓		
GS-agency FE			✓	✓
DV:	# Staff Without Graduate Degree			
	<i>GS Pooled</i>	<i>GS: Mid Level</i>	<i>GS: Upper Level</i>	
	(1)	(2)	(3)	(4)
Demand × Climate-relevant	−1730.490+ (891.089)		−128.567+ (63.088)	−127.432* (50.178)
Demand (overlapping) × Climate-relevant		−1092.672 (635.654)		
Climate-relevant bureaucrat	9080.531 (12518.692)	9216.950 (12510.104)	−1103.723 (945.583)	−442.808 (1008.818)
N	2,276	2,276	9,379	6,591
Adjusted R ²	0.626	0.626	0.636	0.636
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
Agency FE	✓	✓		
GS-agency FE			✓	✓
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

Table G1: Replication of Table 1, replacing proportional outcome with *counts* of bureaucrats with (top panel) and without (bottom panel) a graduate degree. Latter results suggest that the proportional results do not owe to firm demand correlating with growth in the denominator (total number of bureaucrats).

DV:	% Staff With Graduate Degree				
	(1)	<i>GS Pooled</i> (2)	(3)	<i>GS: Mid Level</i> (4)	<i>GS: Upper Level</i> (5)
Demand $\times$ Climate-relevant	−1.233*** (0.309)	−1.272*** (0.320)		−1.377 (1.580)	−2.793** (0.975)
Demand (overlapping) $\times$ Climate-relevant			−0.973*** (0.207)		
Climate-relevant bureaucrat	44.587*** (9.308)	44.719*** (9.460)	44.253*** (9.442)	73.139* (25.578)	62.256** (19.835)
N	2,276	2,276	2,276	9,379	6,591
Adjusted R ²	0.885	0.884	0.884	0.511	0.735
Software demand control	✓	✓	✓	✓	✓
Presidential transition control		✓	✓	✓	✓
Quarter-year FE		✓	✓	✓	✓
Agency FE	✓	✓	✓		
GS-agency FE				✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table G2: Replication of models in Table 1. Model (1) excludes quarter-year fixed effects. Models (2–5) incorporate a control that interacts the climate-relevant binary with an indicator for quarters of presidential transitions (2017Q1 and 2021Q1).

H. BUREAUCRATIC COMPOSITION: RETENTION AND RECRUITMENT

DV:	Quits		Hires	
	(1)	(2)	(3)	(4)
Demand $\times$ Climate-relevant	0.407*** (0.058)	0.178** (0.043)	0.442*** (0.055)	0.156*** (0.062)
Climate-relevant bureaucrat	0.925 (0.800)	1.397* (0.555)	−0.986 (1.014)	−0.073 (0.614)
N	2,200	24,200	2,200	24,200
Adjusted R ²	0.864	0.710	0.849	0.733
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
Agency FE	✓		✓	
GS-agency FE		✓		✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H1: Regressions of logged counts of quits and hires on firm demand in the preceding quarter (logged count of job postings), interacted with the climate relevance of job type. Standard errors clustered by agency.

	<i>GS: Mid Level</i>		<i>GS: Upper Level</i>	
	DV: Quits	DV: Hires	DV: Quits	DV: Hires
	(1)	(2)	(3)	(4)
Demand $\times$ Climate-relevant	0.003 (0.083)	−0.002 (0.117)	0.161* (0.058)	−0.028 (0.104)
Climate-relevant bureaucrat	0.504 (0.704)	−2.297+ (1.272)	2.925** (0.915)	−2.475 (1.557)
N	4,996	5,519	3,568	3,595
Adjusted R ²	0.744	0.748	0.703	0.708
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
GS-agency FE	✓	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H2: Regressions of logged counts of quits and hires of workers **with graduate degrees** on firm demand in the preceding quarter (logged count of job postings), interacted with the climate relevance of job type. Mid-level GS grades are GS-8 through GS-12. Upper-level GS grades are GS-13 and above. Standard errors clustered by agency.

	<i>GS: Mid Level</i>		<i>GS: Upper Level</i>	
	DV: Quits	DV: Hires	DV: Quits	DV: Hires
	(1)	(2)	(3)	(4)
Demand $\times$ Climate-relevant	0.026 (0.069)	0.071 (0.113)	−0.001 (0.070)	−0.093 (0.088)
Climate-relevant bureaucrat	−0.926 (0.859)	−3.151* (1.163)	1.506+ (0.830)	−1.103 (1.285)
N	4,996	5,519	3,568	3,595
Adjusted R ²	0.768	0.760	0.722	0.724
Software demand control	✓	✓	✓	✓
Quarter-year FE	✓	✓	✓	✓
GS-agency FE	✓	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H3: Regressions of logged counts of quits and hires of workers **without graduate degrees** (i.e., a four-year college degree only) on firm demand in the preceding quarter (logged count of job postings), interacted with the climate relevance of job type. Mid-level GS grades are GS-8 through GS-12. Upper-level GS grades are GS-13 and above. Standard errors clustered by agency.

I. CDP GRADES

CDP is a non-profit organization that maintains an environmental disclosure system for firms. CDP solicits disclosures from a broad cross-section of firms around the world on their climate activities

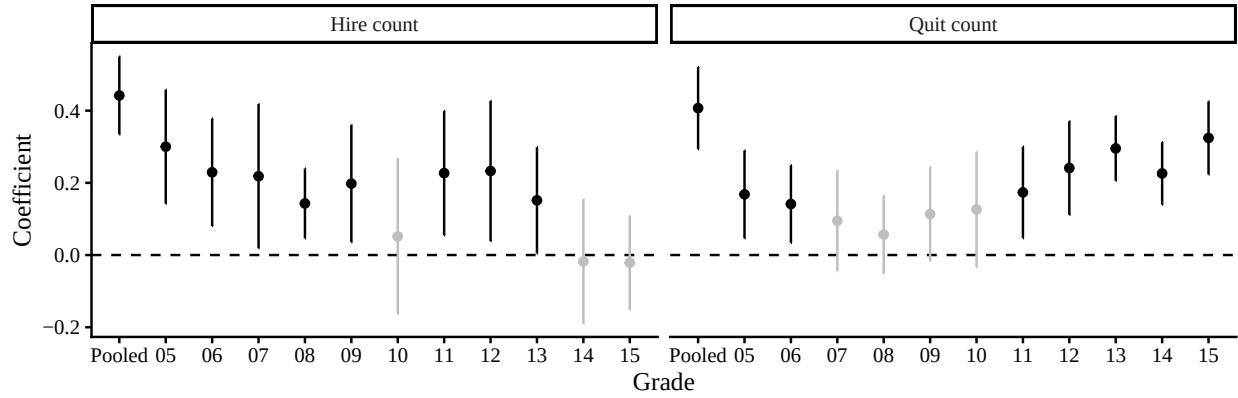


Figure H1: Replication of Figure 3, with hire and quit coefficients plotted separately.

and strategies (Green 2010, *Business and Politics*); nearly 23,000 firms disclose to CDP as of 2024.

CDP develops its list of companies from which to request disclosures in several ways. The primary channel is through CDP’s Capital Market Signatories, made up of asset owners, asset managers, banks, and insurers representing over 25% of all institutional assets globally. These signatories submit their portfolio holdings to CDP. Any firm held by a signatory becomes a requested company and is formally asked to disclose through CDP. Firms can also be selected on the basis of supply-chain membership (if a reporting firm requests disclosures from their other suppliers), the emissions intensity of their sector, and past CDP disclosure history.

Firms respond to a detailed CDP questionnaire that includes questions pertaining to climate governance structures; the identification, assessment, and management of climate-related impacts, risks, and opportunities for the business; the incorporation of climate change in overall firm strategy; environmental performance; and emissions data and methodology, and verification. Based on these firm-level disclosures, CDP awards each firm a grade from A to D that reflects its level of action on climate and the transparency of its climate-related activities and strategies; firms that do not respond to an invitation to disclose climate information receive an F.

1.1. Effect of Firm Demand on Quits, Disaggregating Firms by CDP Grade

	Total Hiring Firms	<i>Non-U.S. HQ</i>	<i>U.S. HQ</i>
A Grade	101	71	30
Sub-A Grade	1,894	798	1,096
% A Grade	5%	8%	3%

Table I1: Counts of unique climate expert-hiring firms by 2014 CDP grade and headquarters location (firms not contacted by CDP are excluded). A-grade firms hire at a rate three times that of sub-A firms. Note that as all job postings are for U.S.-based positions, many of the firms with foreign headquarters are U.S. subsidiaries of foreign parent firms.

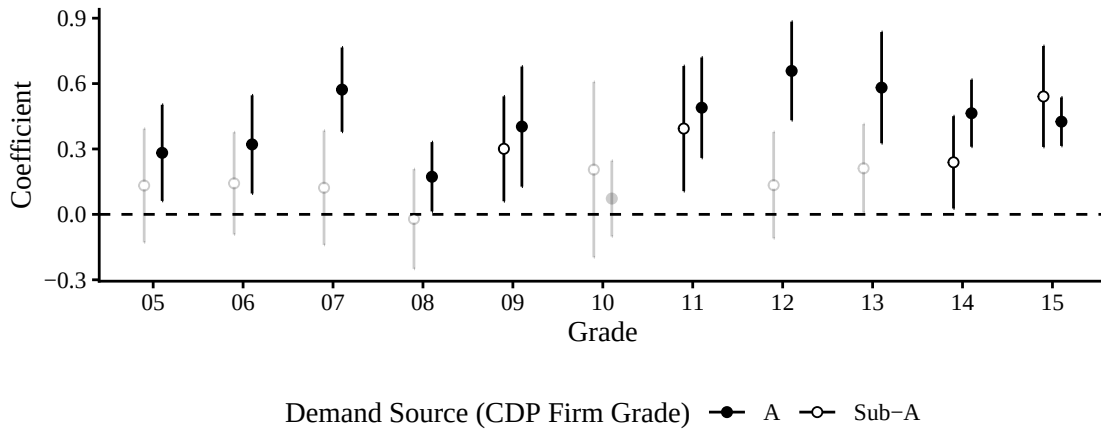


Figure I1: Regressions of logged counts of quits on firm demand in the preceding quarter, interacted with the climate relevance of job type. Demand disaggregated by firms' CDP grade (A, or lower than an A). Sample split by GS grade. Figure plots interaction coefficients of these regressions estimated by individual GS grade; results significant at $p < 0.05$ are darkened. Standard errors clustered by agency.

DV:	Quits		
	<i>GS: Low Level</i> (1)	<i>GS: Mid Level</i> (2)	<i>GS: Upper Level</i> (3)
Demand (A grade) $\times$ Climate-relevant	0.392*** (0.095)	0.359*** (0.081)	0.490*** (0.072)
Demand (sub-A grade) $\times$ Climate-relevant	0.132 (0.102)	0.203* (0.096)	0.330*** (0.081)
Climate-relevant bureaucrat	17.260*** (3.283)	15.536*** (2.979)	20.153*** (2.820)
N	1,440	2,400	1,440
Adjusted R ²	0.679	0.681	0.757
Software demand control	✓	✓	✓
Quarter-year FE	✓	✓	✓
GS-agency FE	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table I2: Regressions of logged counts of quits on firm demand in the preceding quarter (logged count of job postings), interacted with the climate relevance of job type. Demand disaggregated by firms' CDP grade (A, or lower than an A). Sample split across low-level (GS-5 through GS-7), mid-level (GS-8 through GS-12), and upper-level (GS-13 through GS-15) grades. Standard errors clustered by agency.

I.2. Do Firms Improve Their CDP Ratings Upon Hiring Climate Experts?

Sub-A grade firms may seek to improve their climate performance and hire experts in furtherance of this goal. This, if true, would challenge our contention that climate experts at grade-A firms do pro-social work than those at sub-A firms. We conduct a straightforward test of this alternative conjecture: if firms that hire climate experts go on to *improve* their future CDP ratings, this would suggest that even lower-performing firms have pro-social motivations for hiring experts.

DV:	Disclosure (0–100)		Performance (0–5)		Perf. Public (0,1)	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm-level demand (log, $t - 1$)	−0.085 (0.113)	−0.074 (0.110)	−0.016 (0.013)	−0.012 (0.012)	−0.001 (0.002)	−0.001 (0.002)
N	4,005	3,984	3,516	3,503	6,011	5,992
Adj. R ²	0.833	0.839	0.677	0.692	0.828	0.852
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓		✓		✓	
Country-year FE		✓		✓		✓
Industry-year FE		✓		✓		✓

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table I3: Regressions of firm-level CDP grades on firm level demand. Firm-level demand for climate experts has no effect on firms' future CDP ratings.

To conduct this test, we create a panel of firms' annual CDP ratings in 2011, 2012, 2013, and 2014 (the four years for which we also have data on lagged demand for climate experts). We look at three different CDP ratings: strength of emissions disclosure, strength of emissions performance (the letter grade, transformed into an ordinal variable), and an indicator of whether the firm's performance score is public. We estimate fixed effects regressions of the ratings outcomes on firms' logged demand for climate experts in the previous year. As Table I3 shows, we find no evidence that the firms that attempt to hire climate experts subsequently improve performance on any of the metrics provided by CDP. Coefficients are negatively signed, null, and small in magnitude. This suggests that firm demand for climate expertise is not correlated with drives to improve firms' climate performance.

J. CONTRACTING

To test for changes in government contracting, we use data from the U.S. General Services Administration on the near universe of procurement contracts signed by all non-military U.S. executive agencies from 2014–2023. The data, which come from the Federal Procurement Data System and were accessed via SAM.gov, contain detailed metadata on approximately nine million federal contracts, encompassing all significant ($> \$10,000$) purchases of goods and services made by government buyers from private contractors. We use industry codes for contractors to identify contracts in which a federal agency purchased consulting services from the private sector, indicative of the agency’s effort to access external expertise. Consulting, more so than other services, captures a relationship in which the agency is paying to access private sector expertise.

EPA use of private consultants increased rapidly beginning in the late 2010s relative to the federal government average. Consulting accounted for less than a quarter of the agency’s purchases in 2014–2017, but well over a third by 2023. We argue that rather than solely reflecting a general increase in federal agencies’ use of consultants, the disproportionate growth in the EPA’s purchase of consulting services shown partly stems from changes in firm demand for climate experts.

To test this claim, we estimate regression models using the procurement contract data. We use the same measure of private sector demand for climate expertise as in previous analyses, interacted with an indicator for whether the contracting agency is the EPA. We take as the outcome an indicator of whether a given contract involves the purchase of consulting services. We predict a positive sign on the interaction term: as private sector demand for climate experts grows, the EPA should increase its reliance on private sector consultants relative to other agencies. We also consider several placebo outcomes — contracts for construction, logistics, and manufacturing — for which our theory predicts no similar effect. For all outcomes, we estimate the following model:

$$Y_{aits} = \alpha \mathbb{I}(a = \text{EPA}) + \beta \text{demand}_t + \delta [\text{demand}_t \times \mathbb{I}(a = \text{EPA})] + \phi_a + \gamma_t + \iota_s + \epsilon_{aits}$$

where a indexes agencies, i contracts, t quarter-years, and s the U.S. states where a contract was carried out. We include state fixed effects to ensure that results are not driven by, for example, the

DV: contract is for...	Consulting		Construction	Logistics	Manufacturing
	(1)	(2)	(3)	(4)	(5)
Demand $\times$ EPA	0.068*** (0.015)	0.067*** (0.014)	0.003 (0.002)	−0.001** (0.000)	0.003 (0.007)
N	7,547,738	7,547,738	7,547,738	7,547,738	7547738
Adjusted R ²	0.017	0.071	0.021	0.012	0.149
Quarter-Year FE		✓	✓	✓	✓
Agency FE		✓	✓	✓	✓
State FE		✓	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table J1: Regressions of indicators for various contract industries on private sector demand, interacted with an indicator variable equal to 1 when the contracting agency is the EPA. Standard errors clustered by state.

opening or closing of regional offices that may have systematically different procurement needs.

Table J1 presents estimation results with robust standard errors clustered at the state level. Models 1–2 support Hypothesis 2. With and without a demanding set of fixed effects, private sector demand for climate experts is strongly associated with increased use of consulting services by the EPA relative to other agencies. The effect size is substantively meaningful: a standard deviation increase in private sector demand is associated with a 3.3 percentage point increase in the EPA’s use of consultants relative to the rest of the federal bureaucracy. We see no meaningful effect on any placebo outcome, all of which are near zero in magnitude (Models 3–5).

DV: contract is for...	Consulting	Special Study
	(1)	(2)
Demand $\times$ EPA	0.074*** (0.018)	0.003 (0.009)
N	7,453,456	7,556,596
Adjusted R ²	0.069	0.019
Quarter-Year FE	✓	✓
Agency FE	✓	✓
State FE	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table J2: Regressions of indicator variables for various contract categories on private sector demand, interacted with an indicator variable equal to 1 when the contracting agency is the EPA. Model (1) was estimated on a sample that does not include contracts related to special studies. Standard errors clustered by state.

Results for contracting may owe not to changes in firm demand, but rather to coincident shifts in presidential prioritization of climate. Potter (2025, *Journal of Politics*) notes that presidential administrations increase the procurement of policy research from the private sector in prioritized issue areas. To address this alternative explanation, we consider the type of contract that Potter identifies as susceptible to presidential manipulation: the commissioning of “special studies.” As reported in Table J2, private sector demand for climate expertise has no statistically or substantively significant relationship with contracting of special studies (Model 2), while the main results for consulting are robust to excluding all special studies from the sample (Model 1). This suggests that change in the EPA’s use of consultants is not a byproduct of shifts in presidential priorities.

What are the consequences of this observed privatization of climate governance? We offer two conjectures. First, as Hart (2003, *Economic Journal*) observes, a government’s decision to contract out services as opposed to providing them internally mirrors the hold-up problem explored by economic theories of the firm. As private contractors specialize in government services, narrowing their pool of potential clients, the government’s ability to extract a lower price for their services increases; contractors may thus refrain from making investments sought by the government.

Second, Hart, Shleifer, and Vishny (1997, *Quarterly Journal of Economics*) model privatization as a choice that delivers cost savings at the risk of diminished service quality. In their model, privatization is more favorable when the government wants to encourage innovation and when the undesirable consequences of cost cutting are obvious enough *ex ante* that they can be prevented by a well-specified contract. The complexities of climate change make it difficult to fully specify a contract that would prevent a consultant from delivering a cost-efficient but welfare-suboptimal policy solution. For example, “climate risk services” firms offer climate models of variable rigor and difficult-to-verify quality (Condon 2023, *Arizona State Law Journal*).

K. GOVERNMENT RESEARCH: ADDITIONAL RESULTS

The top five federal agencies, by count of climate works in analyzed OpenAlex data (aggregating sub-agencies with at least ten works in OpenAlex), are Agriculture, Interior, Commerce, EPA, and Health and Human Services. Agriculture houses the U.S. Forest Service and a number of research

	Citations (log)			
	<i>Demand lag: 2 years</i> (1)	<i>Demand lag: 2 years</i> (2)	<i>Demand lag: 5 years</i> (3)	<i>Demand lag: 5 years</i> (4)
Demand × climate relevance	−0.132** (0.049)		−0.192+ (0.106)	
Demand × government authorship		−0.144*** (0.032)		−0.258* (0.107)
Climate relevance	1.857* (0.809)		3.496** (1.252)	
Government authorship		1.981* (0.728)		3.774*** (0.996)
Sample	Government research		Climate research	
N	43,705	69,794	32,923	57,280
Adjusted R ²	0.150	0.123	0.148	0.121
Software demand control	✓	✓	✓	✓
Quarter-year-field FE	✓	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table K1: Replication of Table 2, lagging firm demand by two or five years.

stations that focus on sustainable land management (e.g., the Rocky Mountain Research Station). Interior houses the U.S. Geological Survey, among other subagencies. Commerce includes the National Oceanic and Atmospheric Administration, a primary climate research body. Health and Human Services encompasses, among other sub-agencies, the Food and Drug Administration and National Institutes of Health. Table K2, shows that proportion of upper-level bureaucrats who hold a graduate degree positively predicts citations to work published by a given agency.

	Citations to climate works (log)
% upper-level staff with graduate degree	3.124+ (1.457)
N	621
Adjusted R ²	0.326
Quarter-year FE	✓
Agency FE	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table K2: Regression of the log count of citations to works published in a given year, by researchers at a given federal agency, on the percentage of upper-level bureaucrats in that agency who hold a graduate degree GS grades 13–15, lagged by one year). Standard error clustered by agency.

L. REGULATORY ENFORCEMENT: ADDITIONAL RESULTS

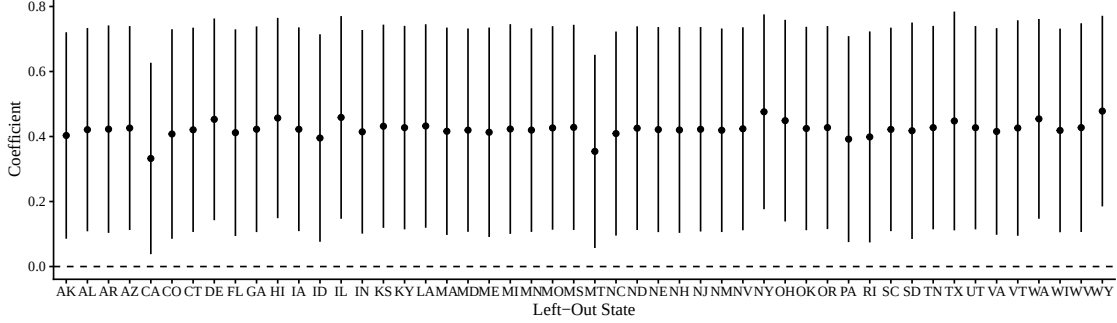


Figure L1: Interaction coefficients from Table 3, reestimating the Model 1 iteratively excluding each state from the sample. Lines depict 95% confidence intervals.

M. IDENTIFICATION

For our primary analyses, we employ a difference-in-differences research design with continuous treatment. The treatment is private sector demand for climate expertise, which varies only at the quarter-year level; the treated group is the set of all climate-relevant workers in federal agencies, while the comparison group is the set of all other federal bureaucrats in those agencies. The variety of causal estimands that can be identified in this type of research design is a topic of active discussion among econometricians; we follow Callaway, Goodman-Bacon, and Sant’Anna (2026, NBER Working Paper 32117; arxiv.org/pdf/2107.02637), hereafter CGS, in defining our estimand as the ATT^o and discuss the necessary assumptions for identification.

First, CGS define the $ATT(d|d)$ as follows:

$$ATT(d|d) = \mathbb{E}[\Delta Y|D = d] - \mathbb{E}[\Delta Y|D = 0]$$

Intuitively, this is the difference between the expected change in the outcome for units with a treatment level of d and the expected change in the outcome of the untreated group. Further, aggregating over all levels of treatment d , we have:

$$ATT^o = \mathbb{E}[\Delta Y|D > 0] - \mathbb{E}[\Delta Y|D = 0]$$

CGS demonstrate that one desirable property of the ATT^o is that it is identified under the “standard” difference-in-differences assumptions: (1) no anticipation, and (2) parallel trends. Both assumptions are plausibly met in our setting. Anticipation would require individual federal bureaucrats to anticipate future change in private sector labor markets. The only plausible case in which this might occur would be presidential transitions; staff might, upon witnessing the election of Biden in 2020, assume that his pro-climate administration would create legislation that would necessitate firm hiring of climate experts (or assume the opposite for Trump in 2016). However, we demonstrate that our results are not driven by presidential transitions (see Table G2).

Empirically assessing the plausibility of the parallel trends assumption is difficult in our setting due to the lack of a “pre-treatment” period. Because our data does not extend back to a period in which there was zero private sector demand for climate expertise, we cannot conduct standard tests of pre-trends. We nonetheless argue that the parallel trends assumption is plausibly met in our context after conditioning on observables and fixed effects. The assumption would be violated if the demand shocks we estimate were correlated with an unobserved time-varying factor that differentially affects the employment outcomes of climate-related vs. non-climate-related bureaucrats. One such likely factor is presidential transition, our results are robust to controlling for this (see Table G2). Another plausible factor is structural shifts in the labor market for high-skilled service workers; we control for this using demand for software engineers.